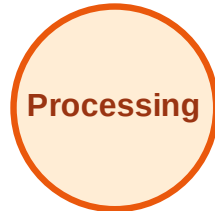


BFS Traversal

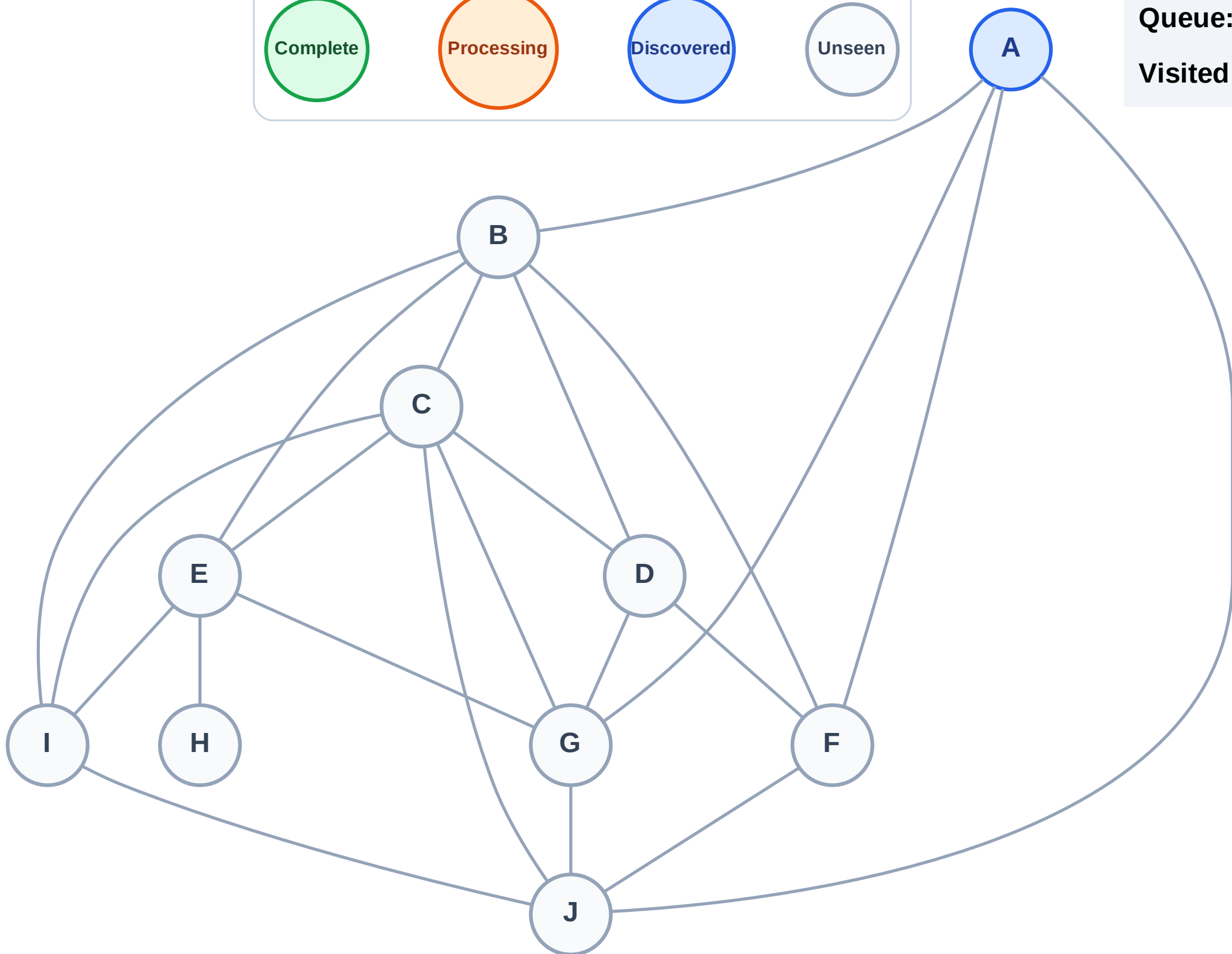
Step 1: Start at A

Legend



Queue: A

Visited: (none)



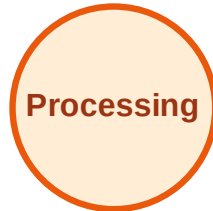
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



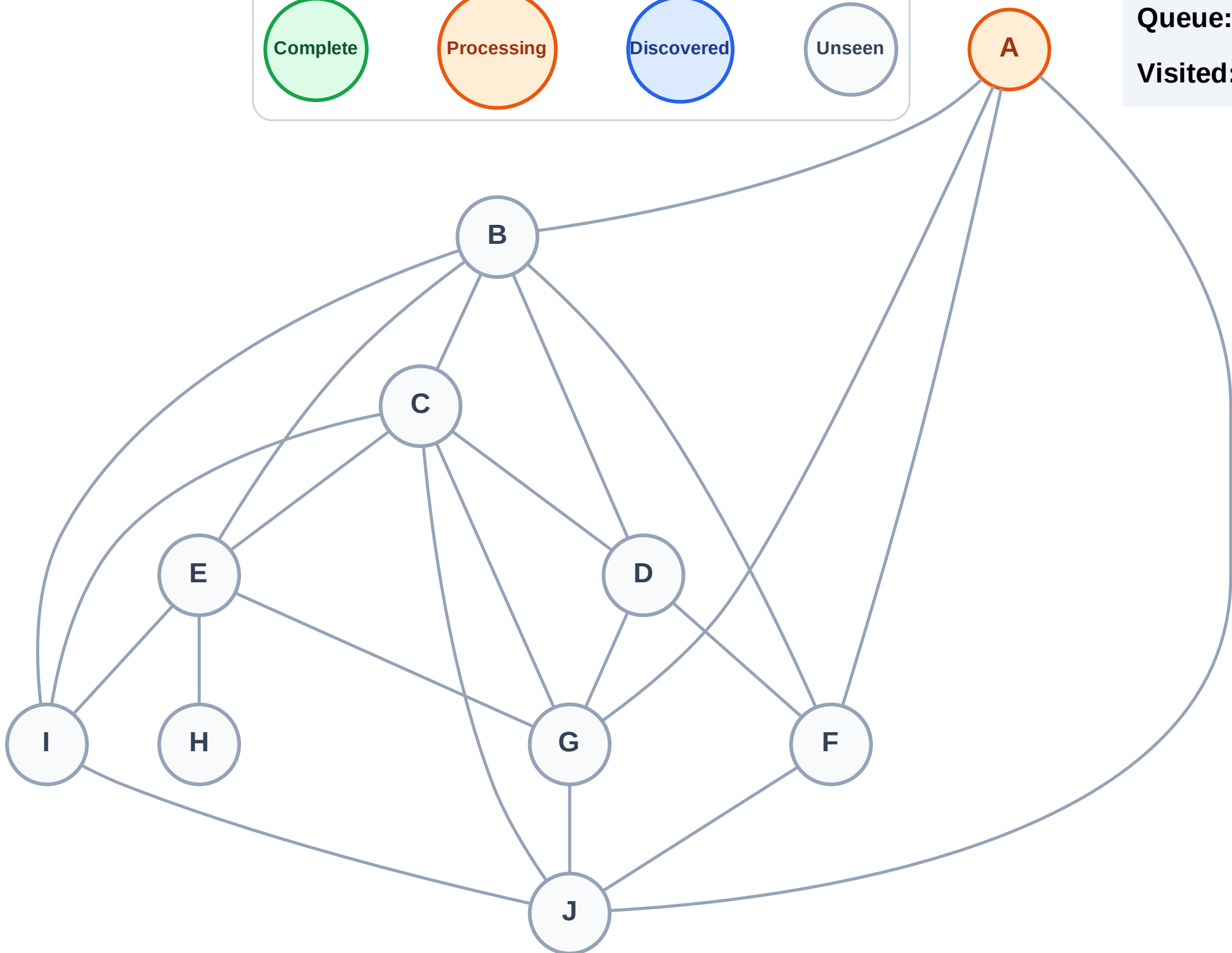
Discovered



Unseen

Queue: (empty)

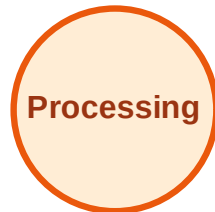
Visited: (none)



BFS Traversal

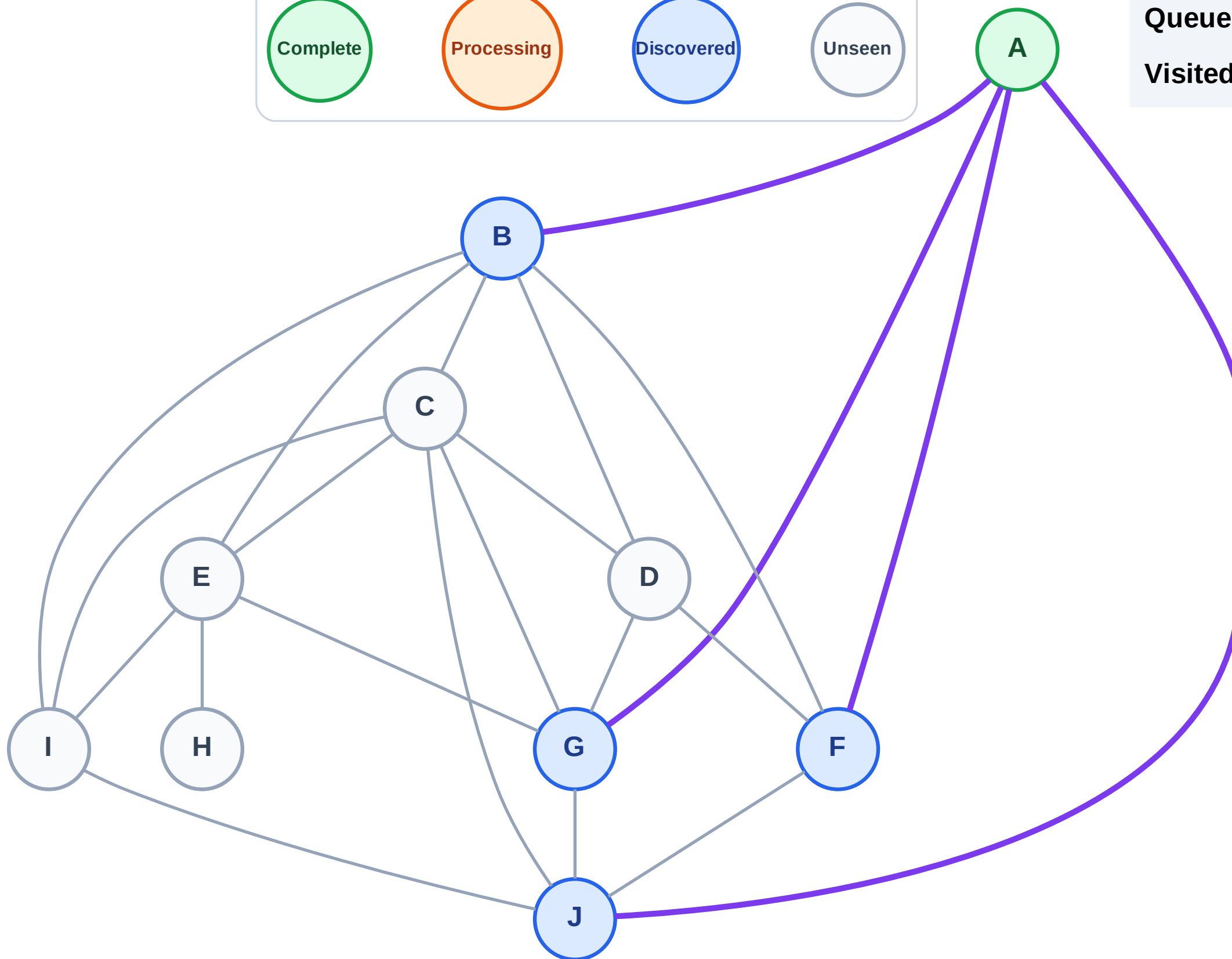
Step 3: Discover B, F, G, J from A

Legend



Queue: B → F → G → J

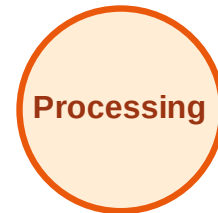
Visited: A



BFS Traversal

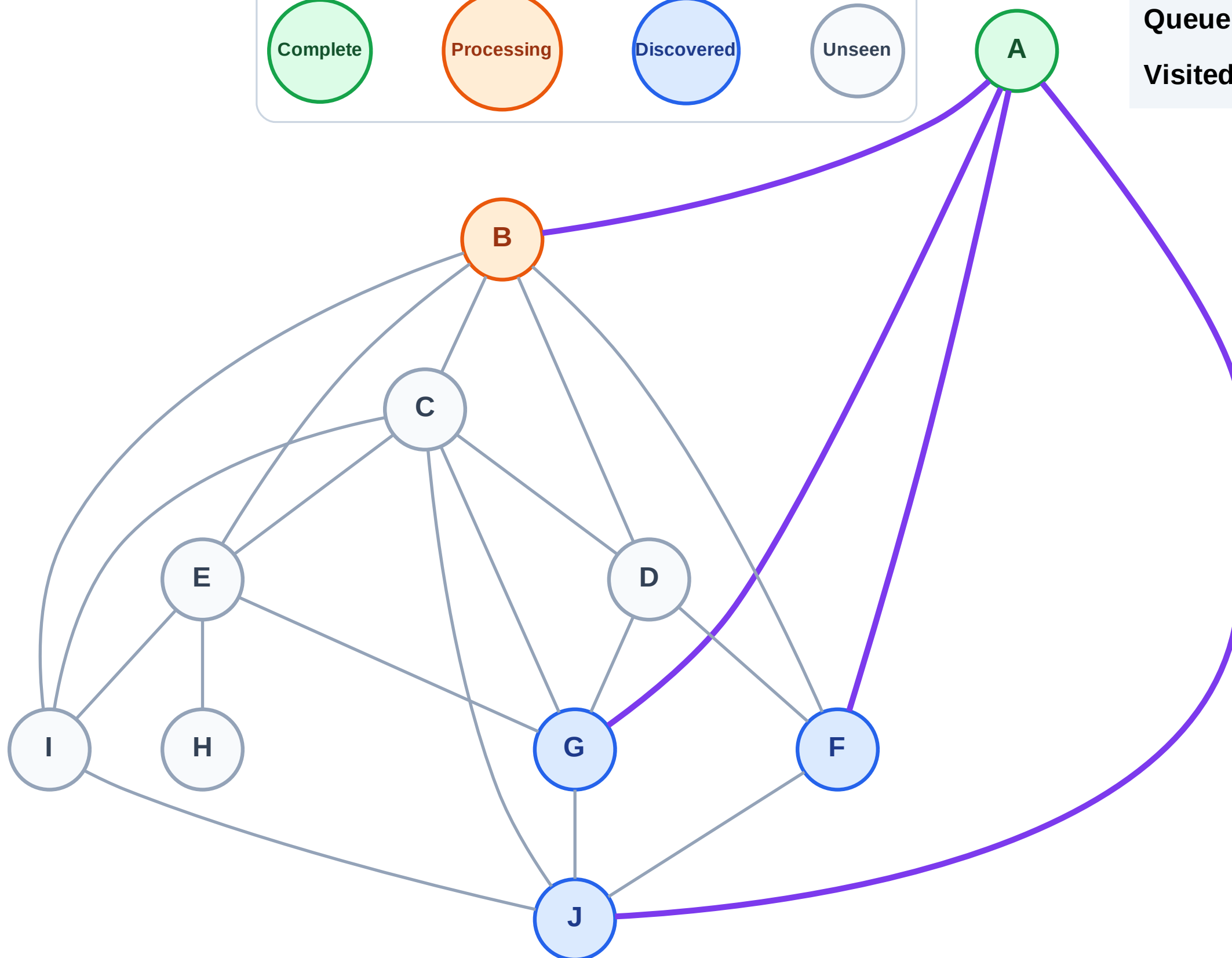
Step 4: Process node B

Legend



Queue: F → G → J

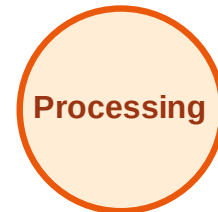
Visited: A



BFS Traversal

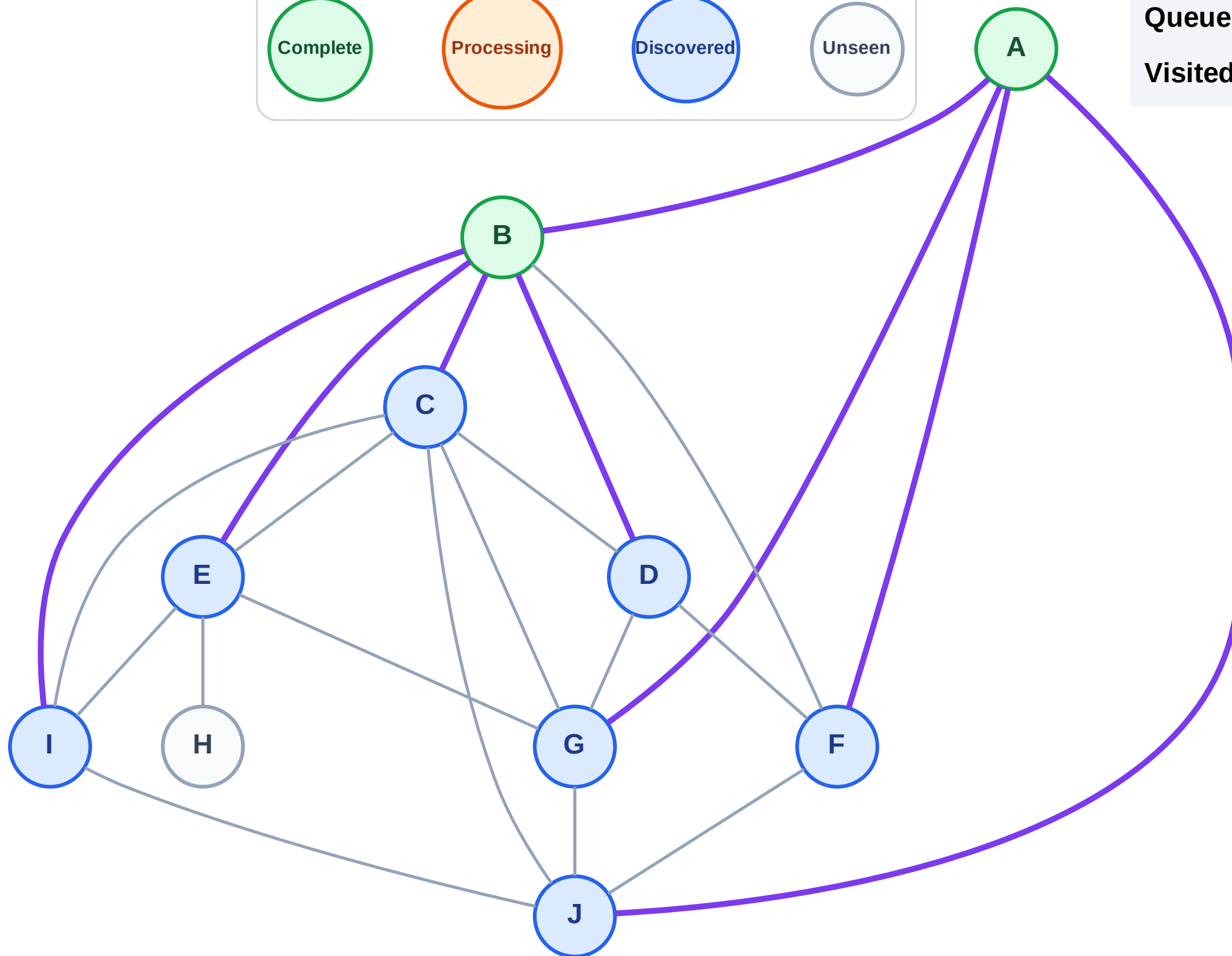
Step 5: Discover C, D, E, I from B

Legend



Queue: F → G → J → C → D → E → I

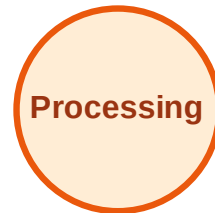
Visited: A, B



BFS Traversal

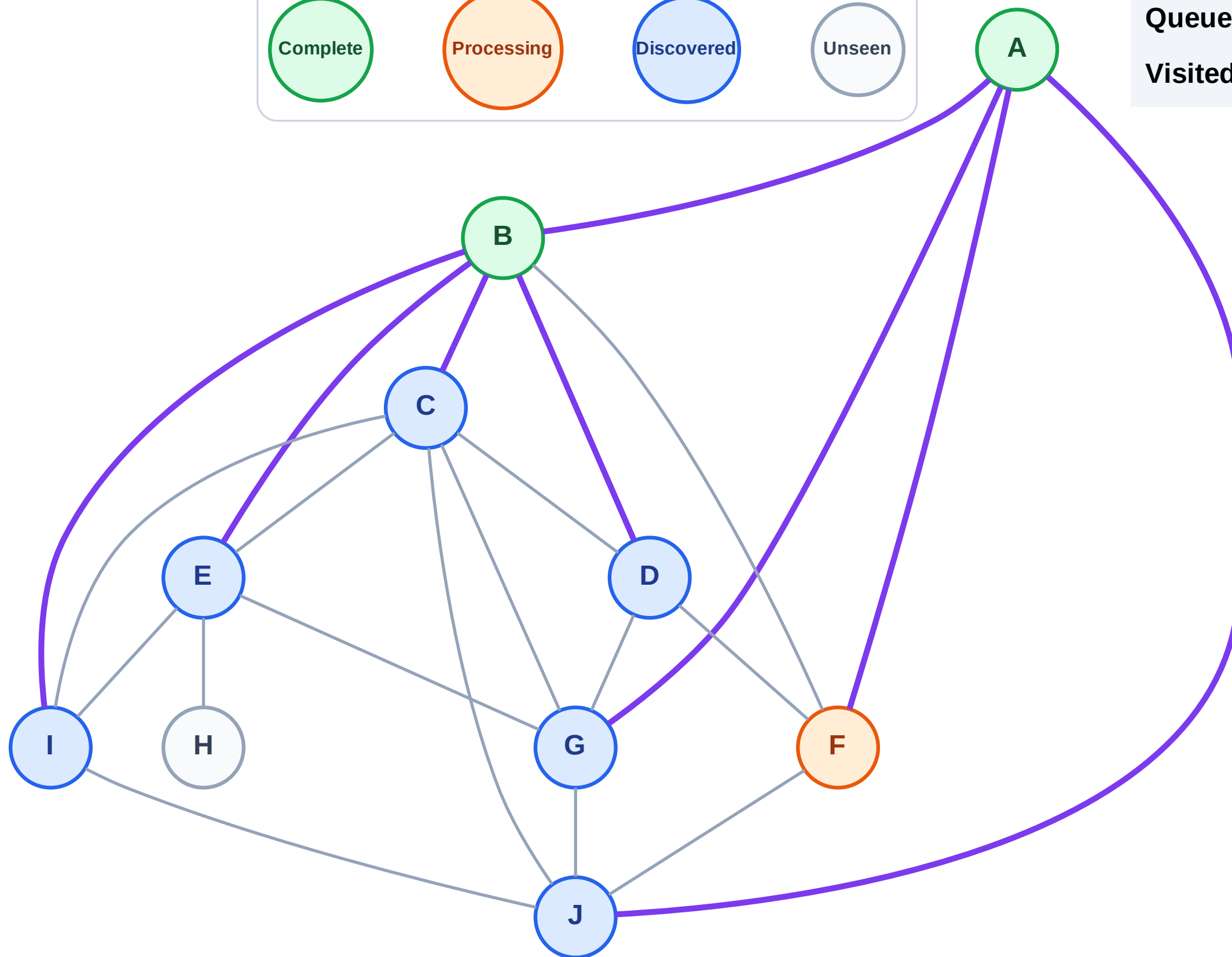
Step 6: Process node F

Legend



Queue: G → J → C → D → E → I

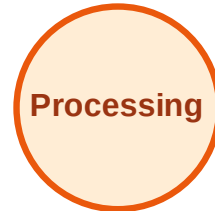
Visited: A, B



BFS Traversal

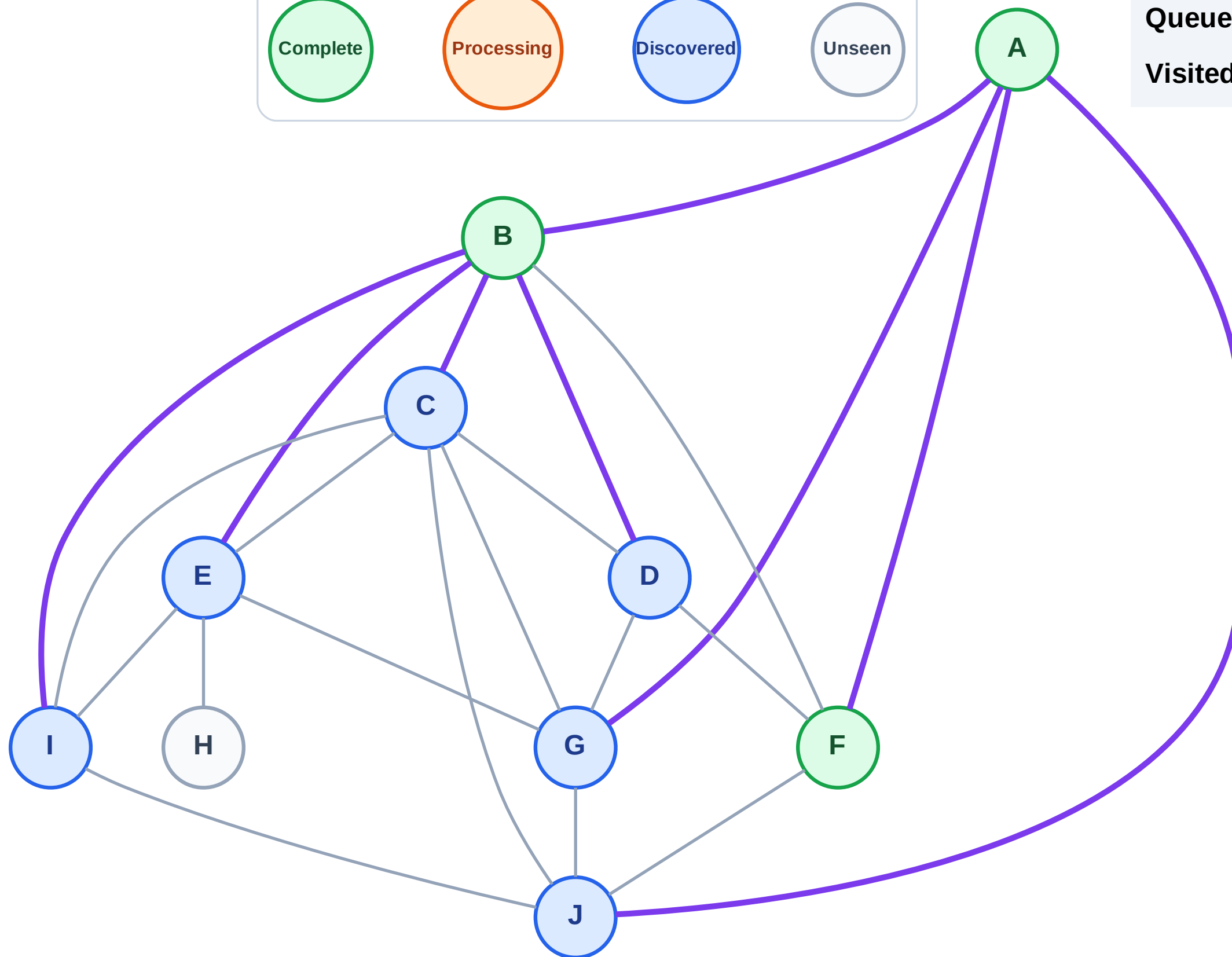
Step 7: No new neighbors from F

Legend



Queue: G → J → C → D → E → I

Visited: A, B, F



BFS Traversal

Step 8: Process node G

Legend

Complete

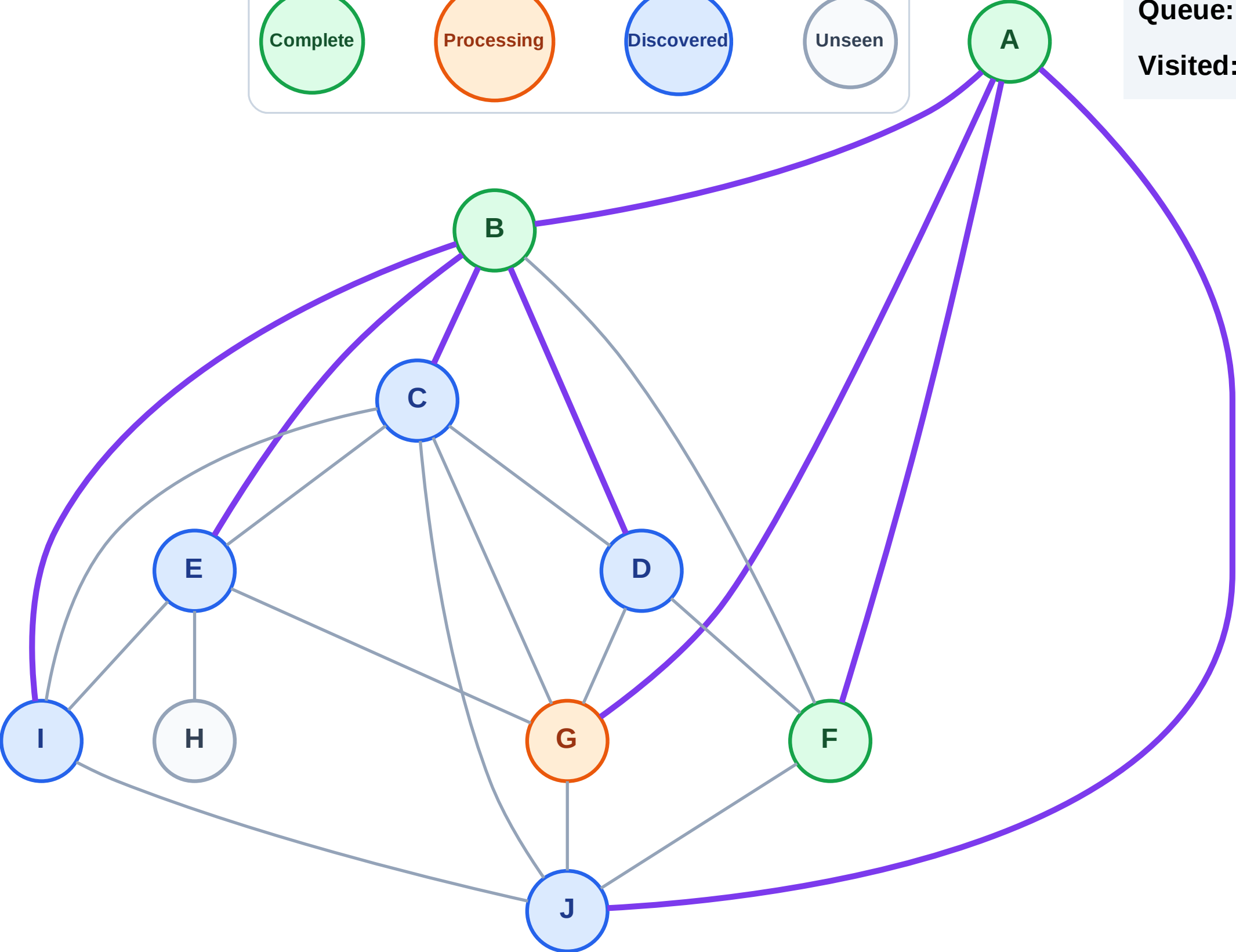
Processing

Discovered

Unseen

Queue: J → C → D → E → I

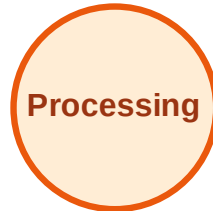
Visited: A, B, F



BFS Traversal

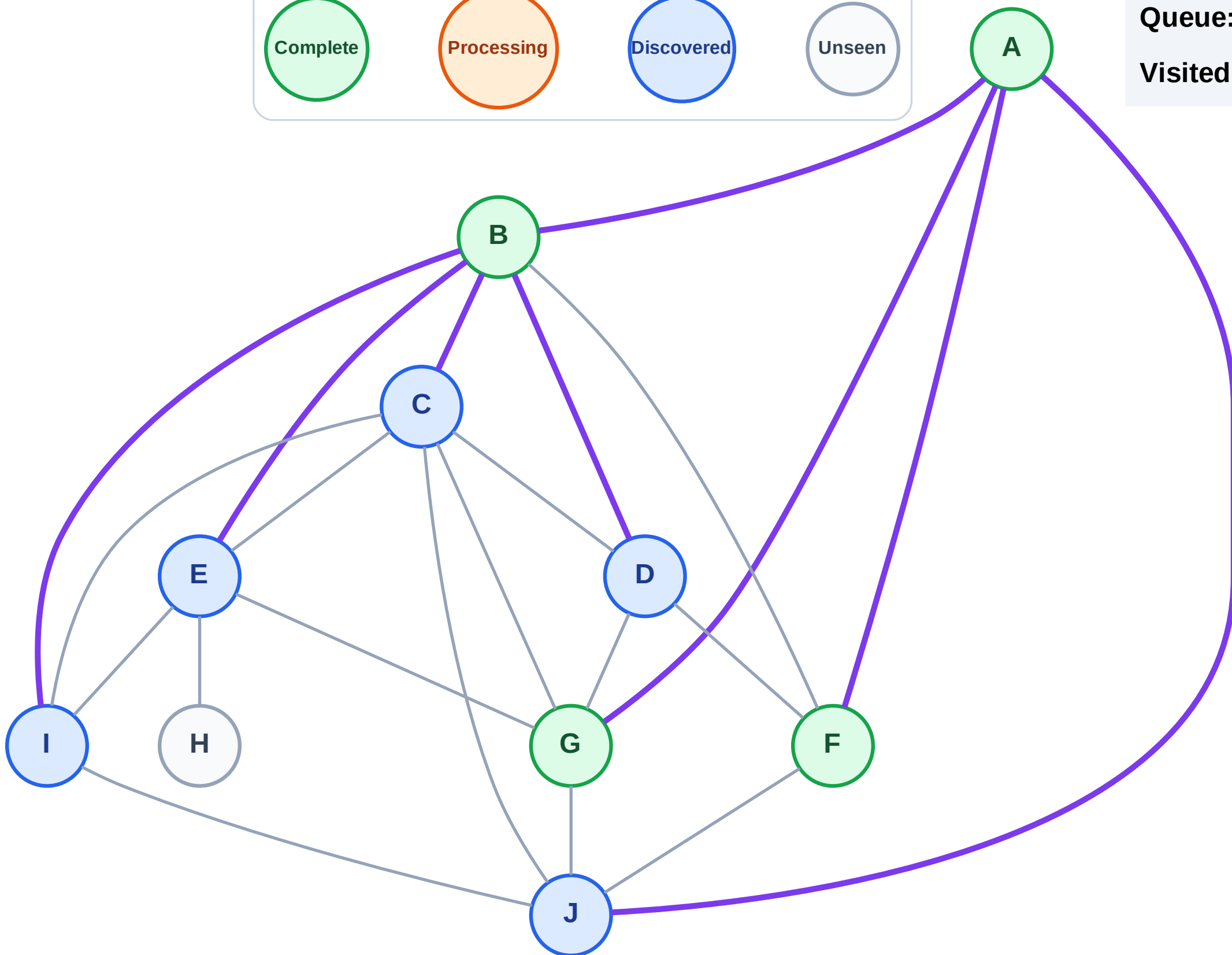
Step 9: No new neighbors from G

Legend



Queue: J → C → D → E → I

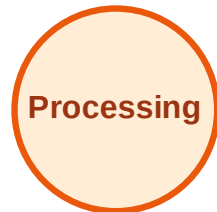
Visited: A, B, F, G



BFS Traversal

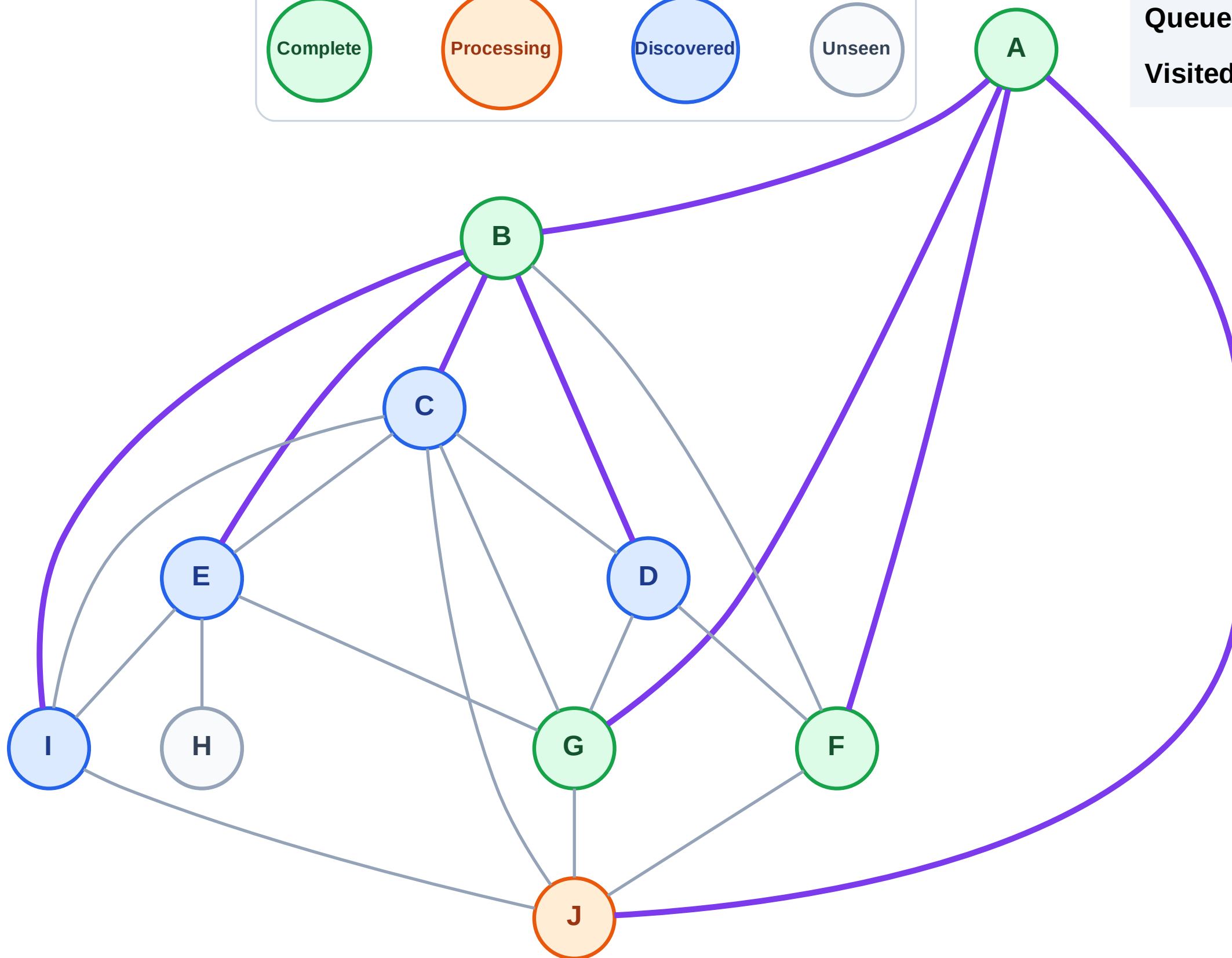
Step 10: Process node J

Legend



Queue: C → D → E → I

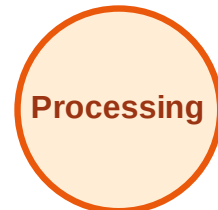
Visited: A, B, F, G



BFS Traversal

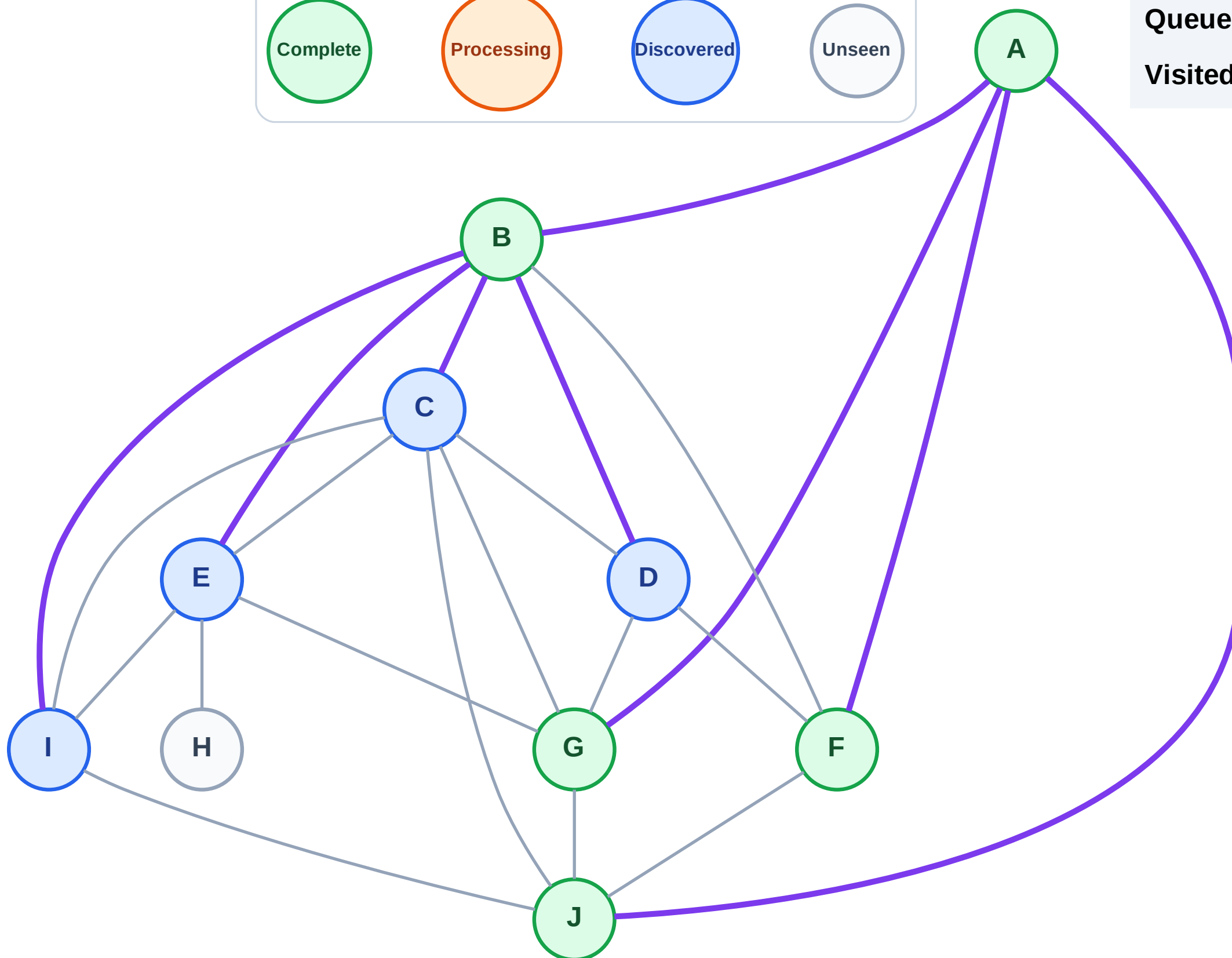
Step 11: No new neighbors from J

Legend



Queue: C → D → E → I

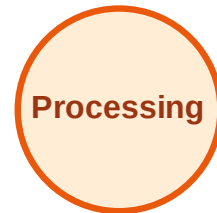
Visited: A, B, F, G, J



BFS Traversal

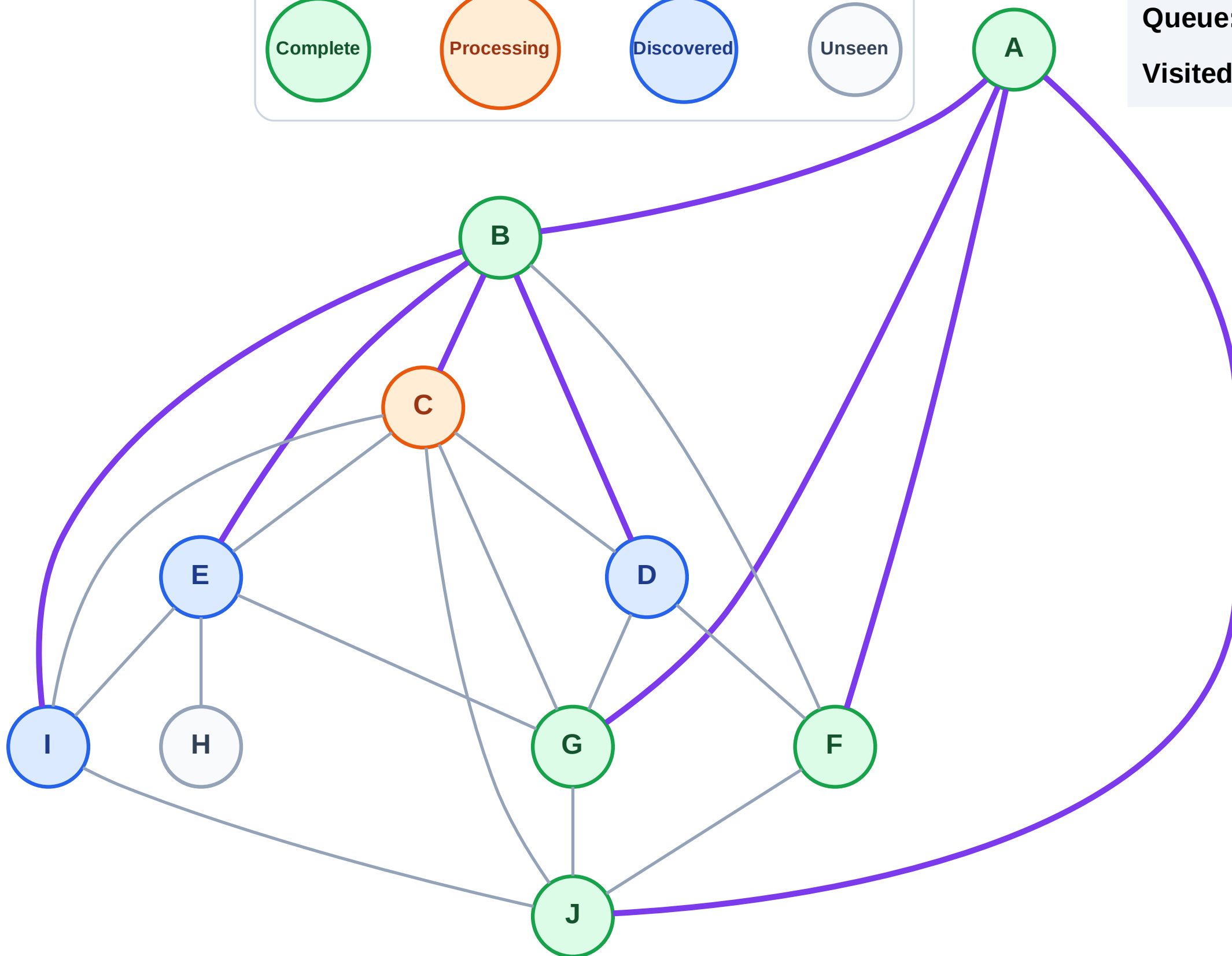
Step 12: Process node C

Legend



Queue: D → E → I

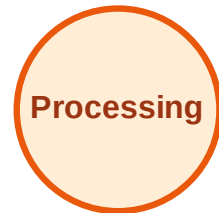
Visited: A, B, F, G, J



BFS Traversal

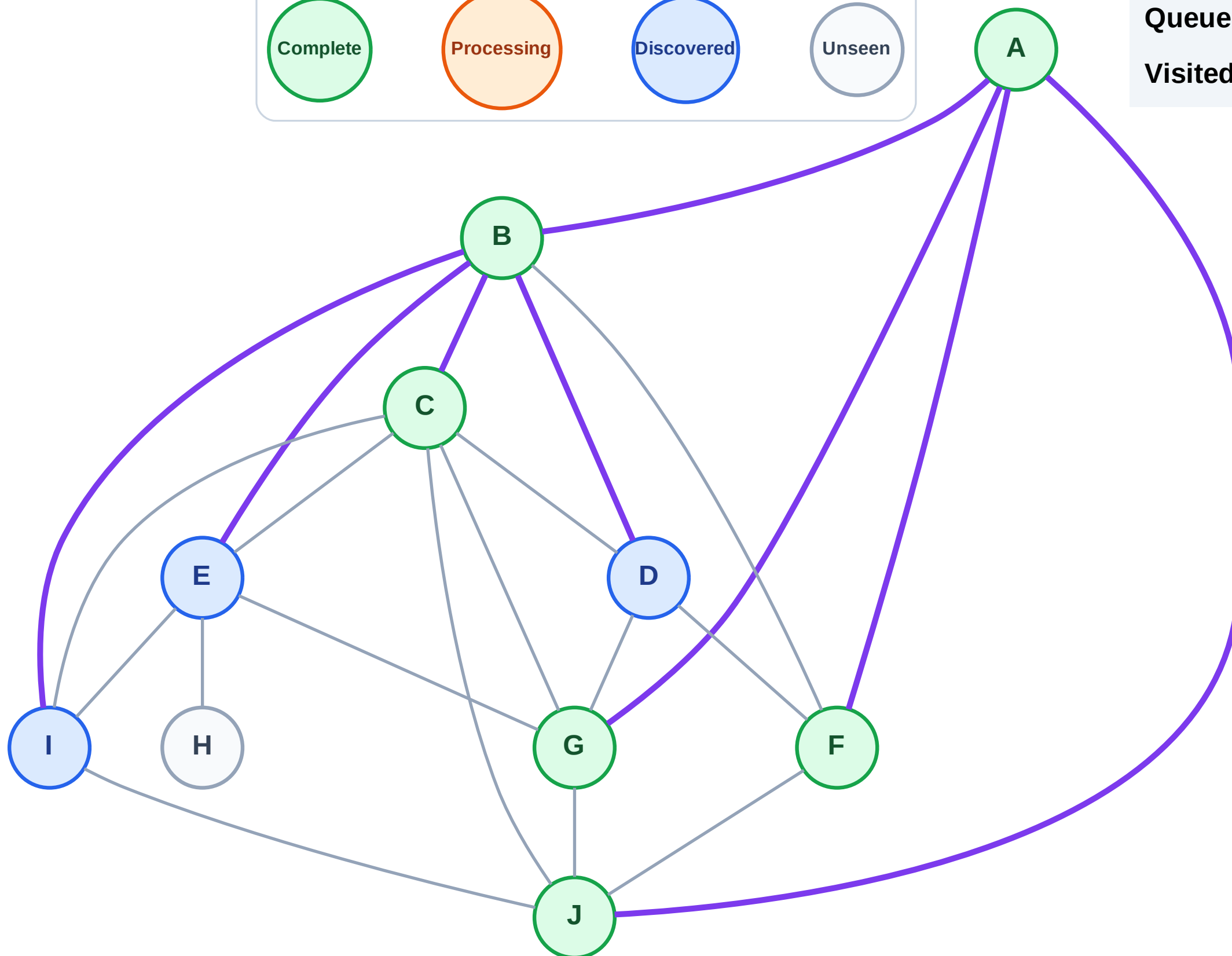
Step 13: No new neighbors from C

Legend



Queue: D → E → I

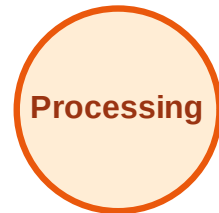
Visited: A, B, C, F, G, J



BFS Traversal

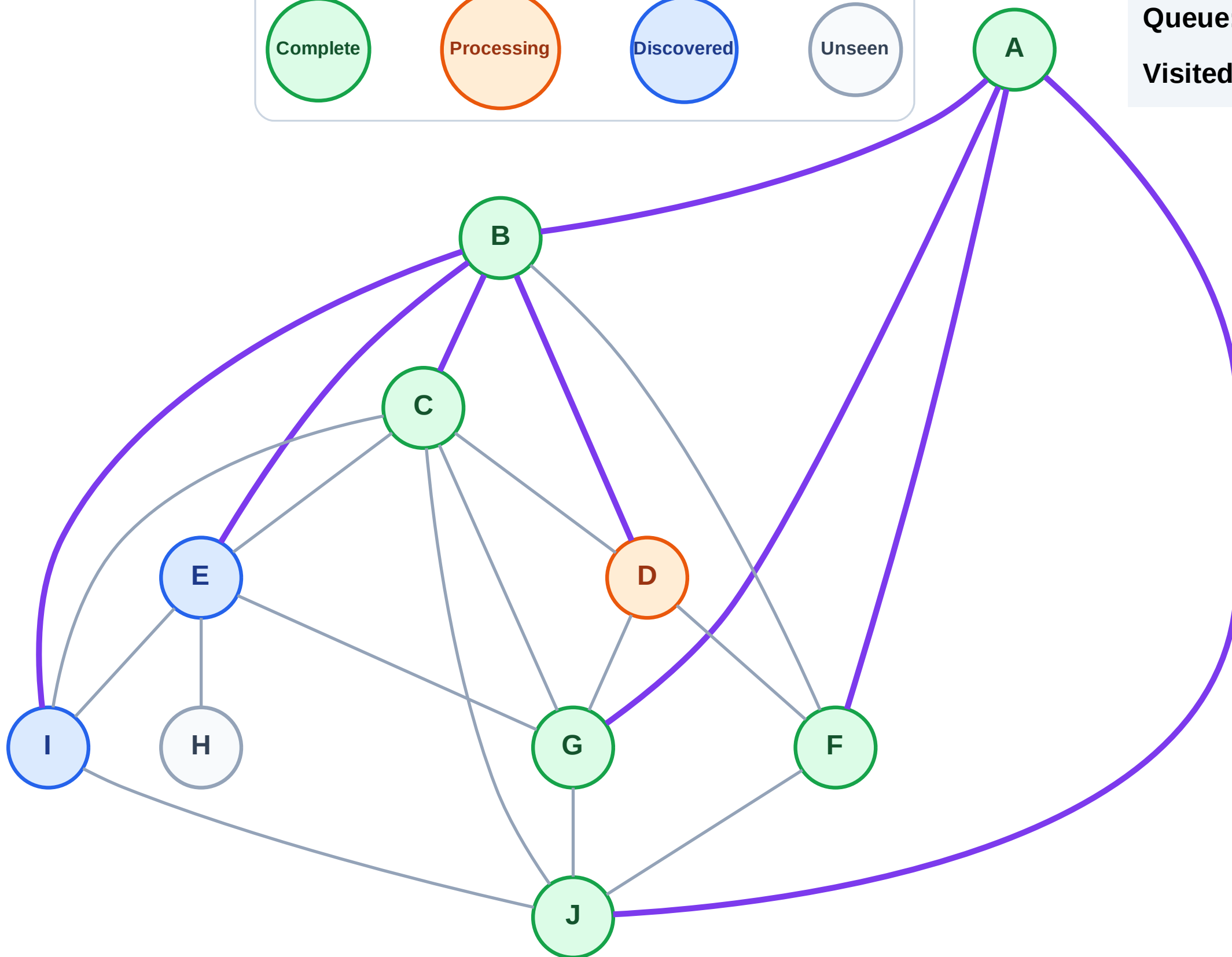
Step 14: Process node D

Legend



Queue: E → I

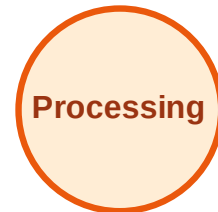
Visited: A, B, C, F, G, J



BFS Traversal

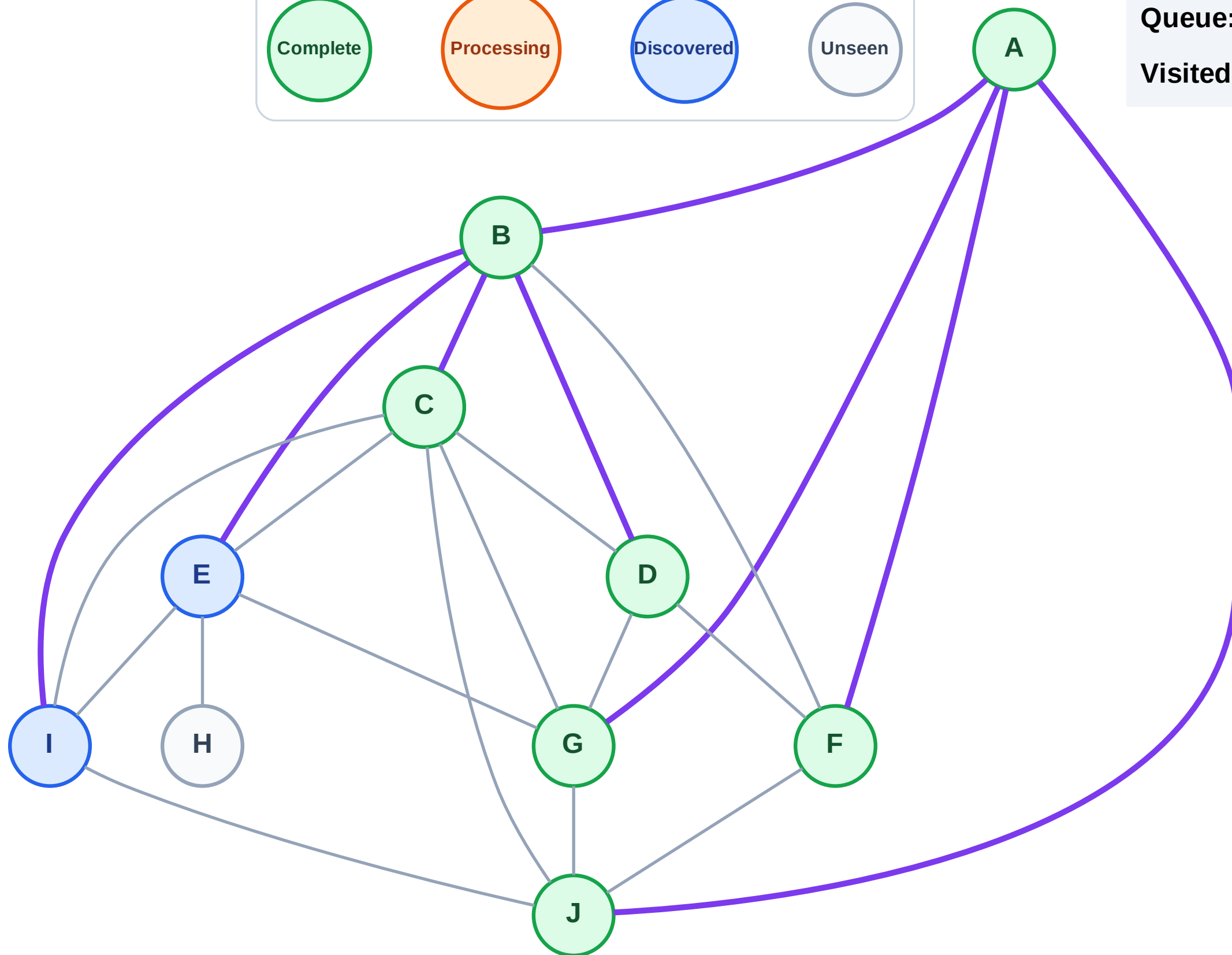
Step 15: No new neighbors from D

Legend



Queue: E → I

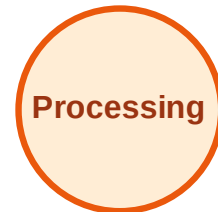
Visited: A, B, C, D, F, G, J



BFS Traversal

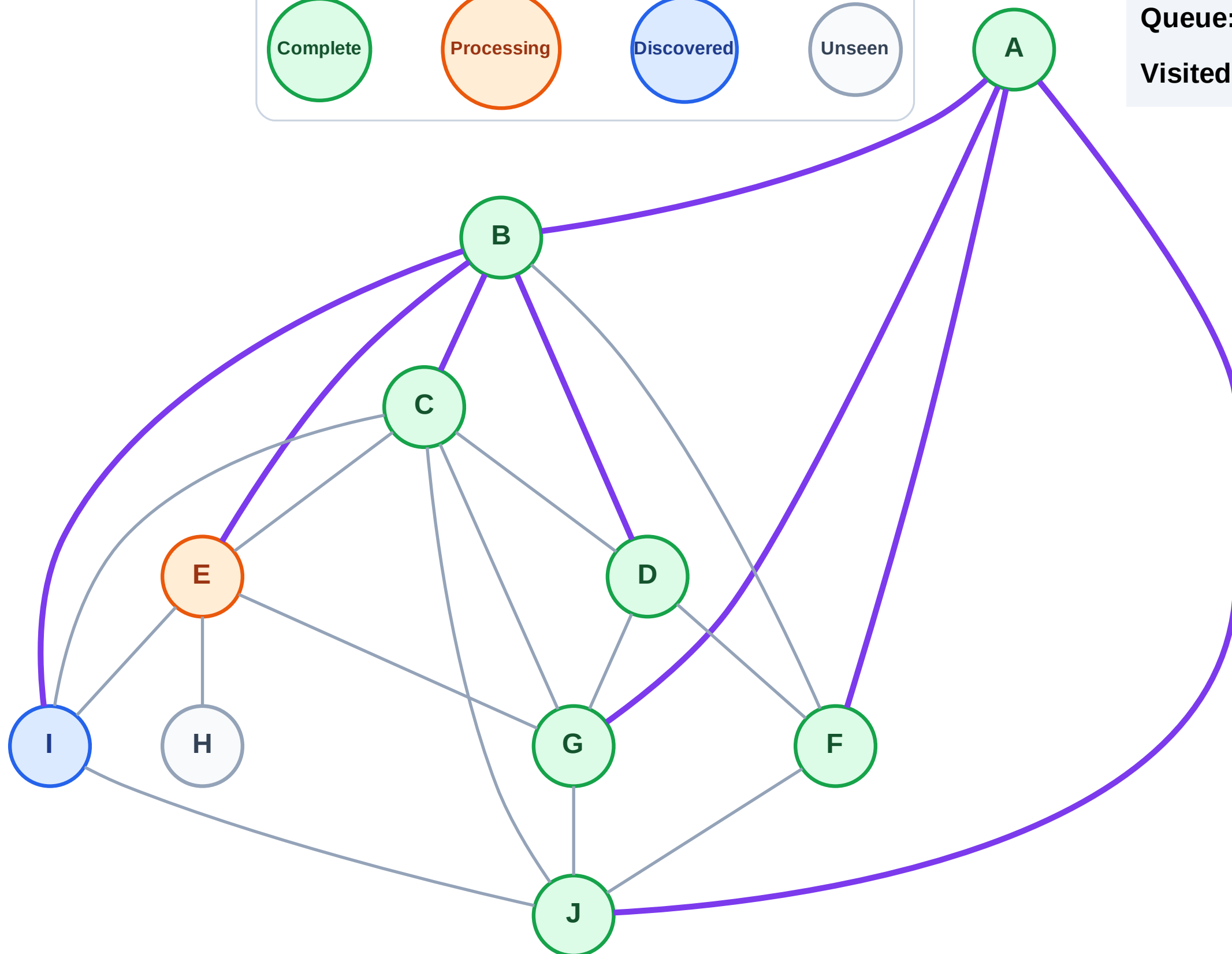
Step 16: Process node E

Legend



Queue: I

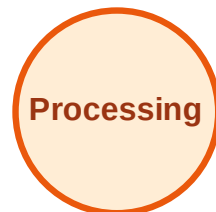
Visited: A, B, C, D, F, G, J



BFS Traversal

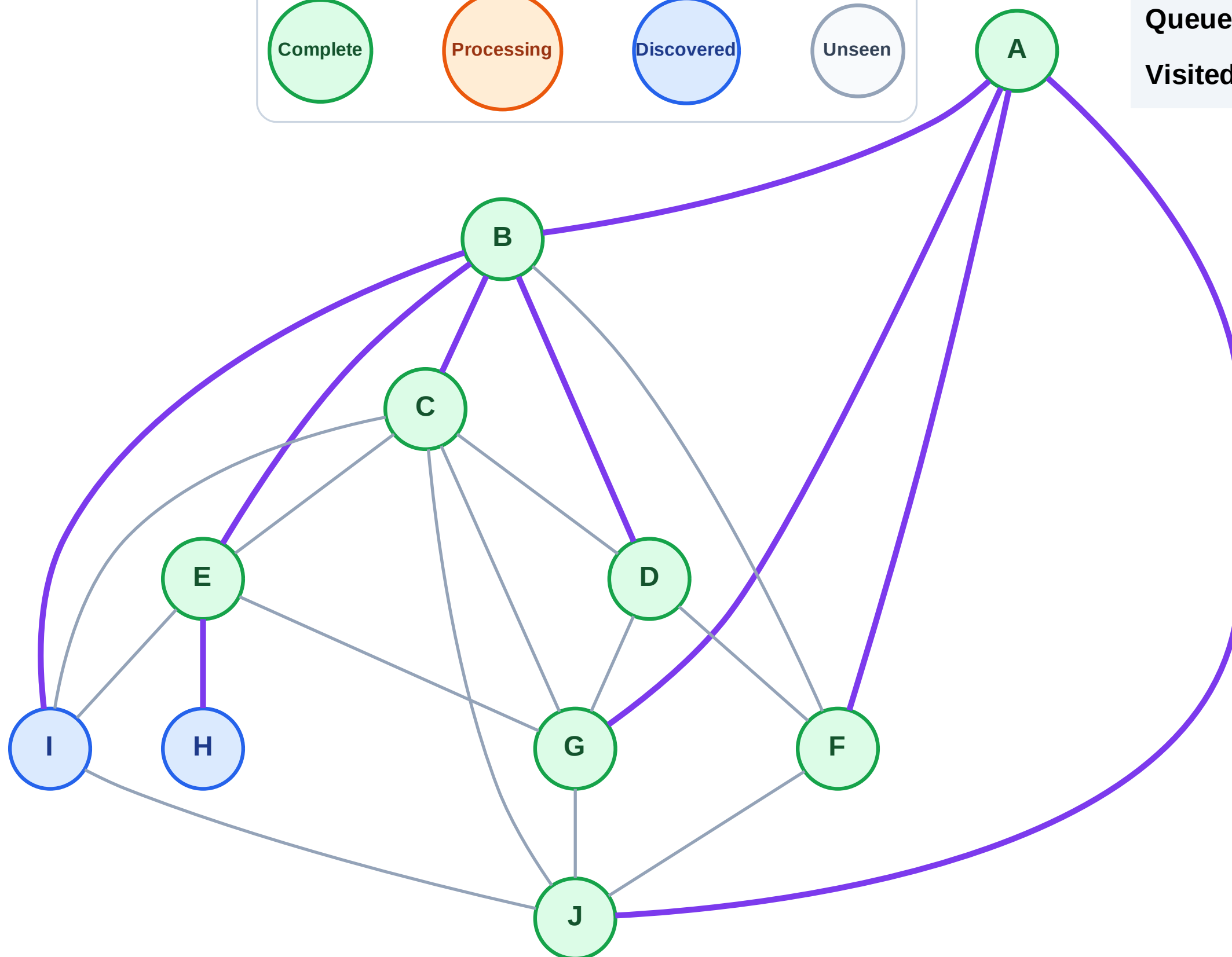
Step 17: Discover H from E

Legend



Queue: I → H

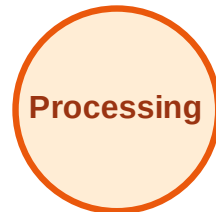
Visited: A, B, C, D, E, F, G, J



BFS Traversal

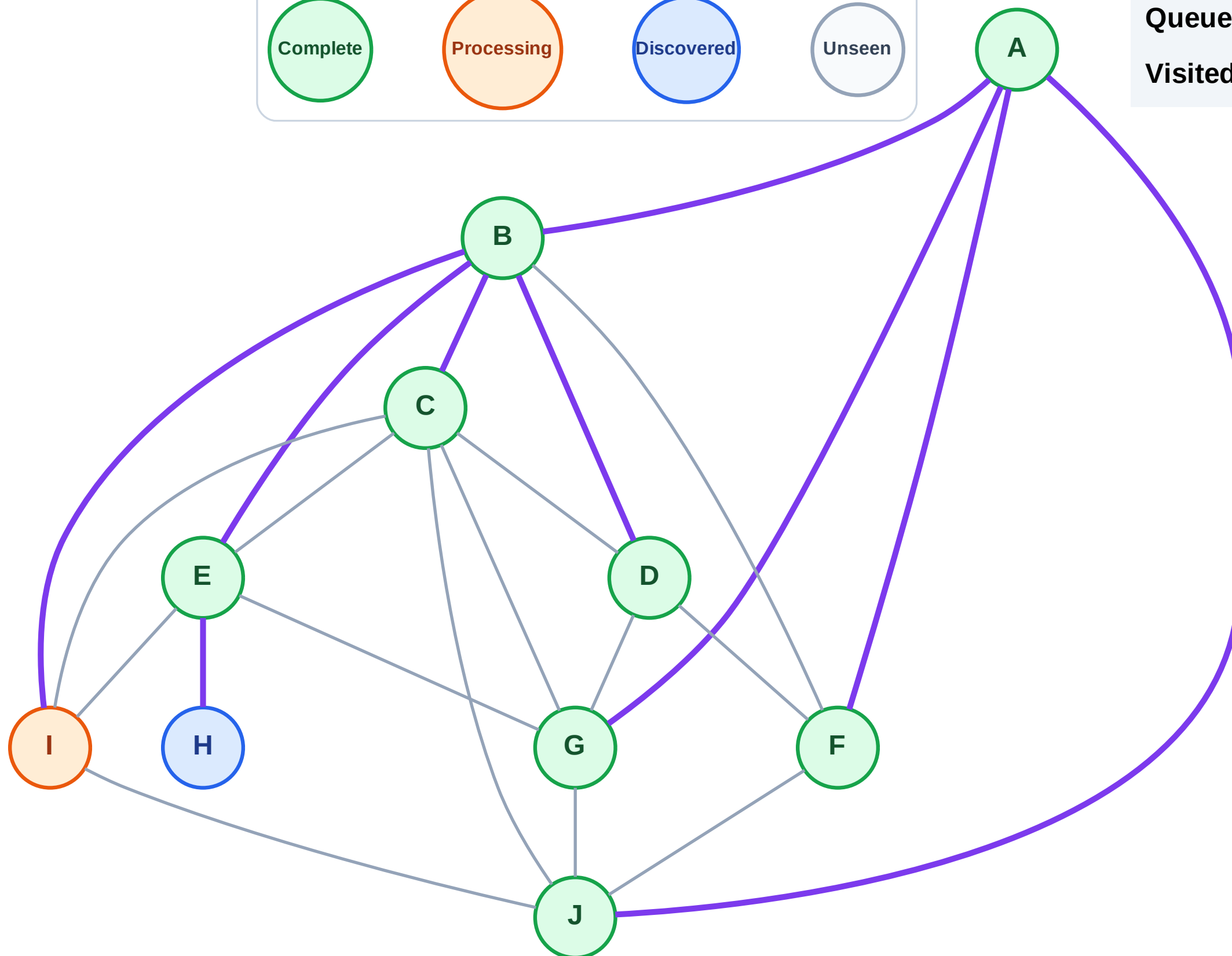
Step 18: Process node I

Legend



Queue: H

Visited: A, B, C, D, E, F, G, J



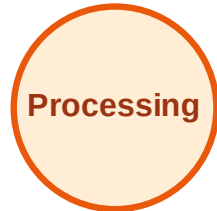
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



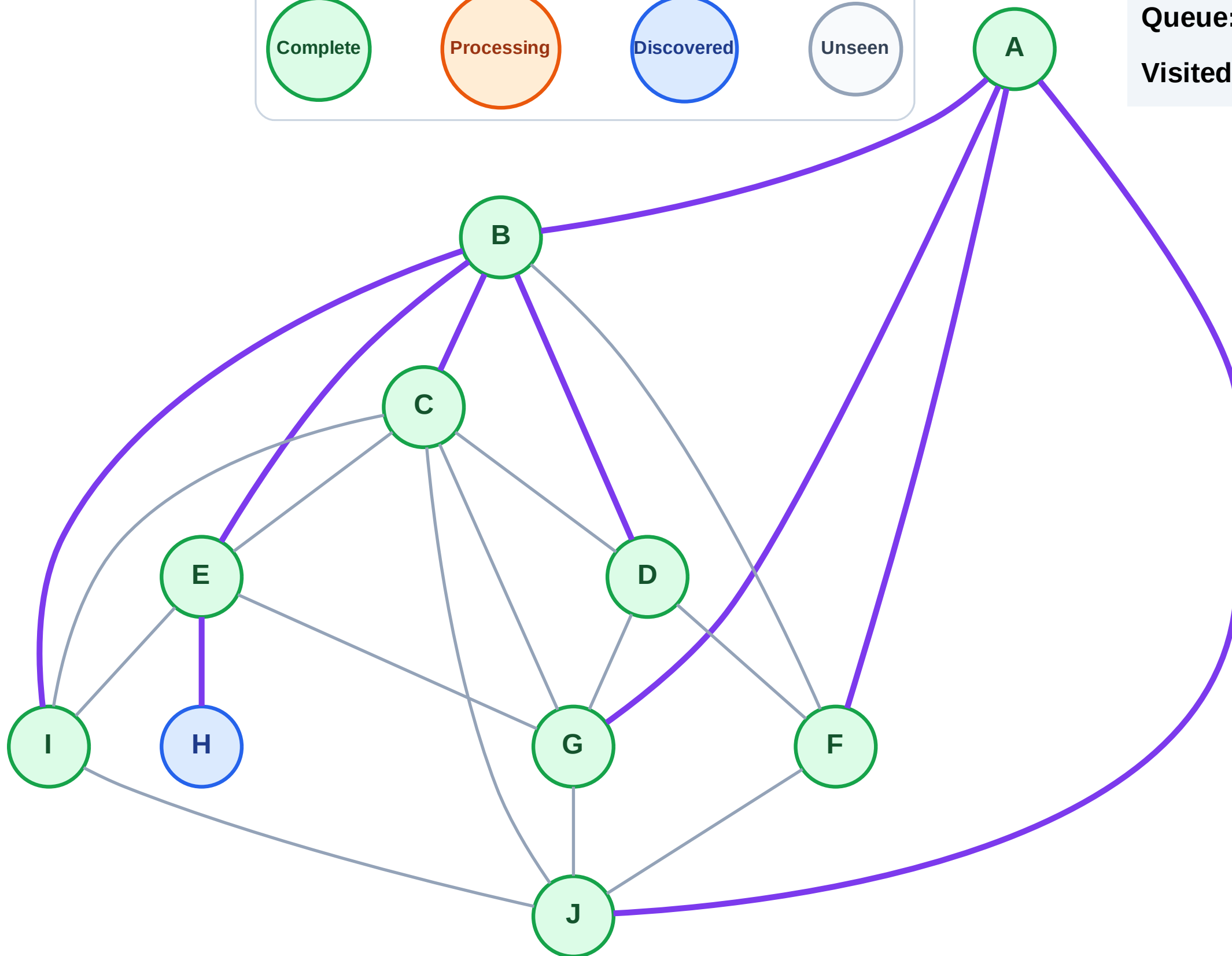
Discovered



Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I, J



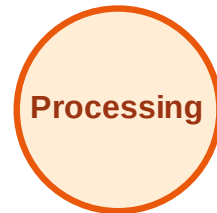
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



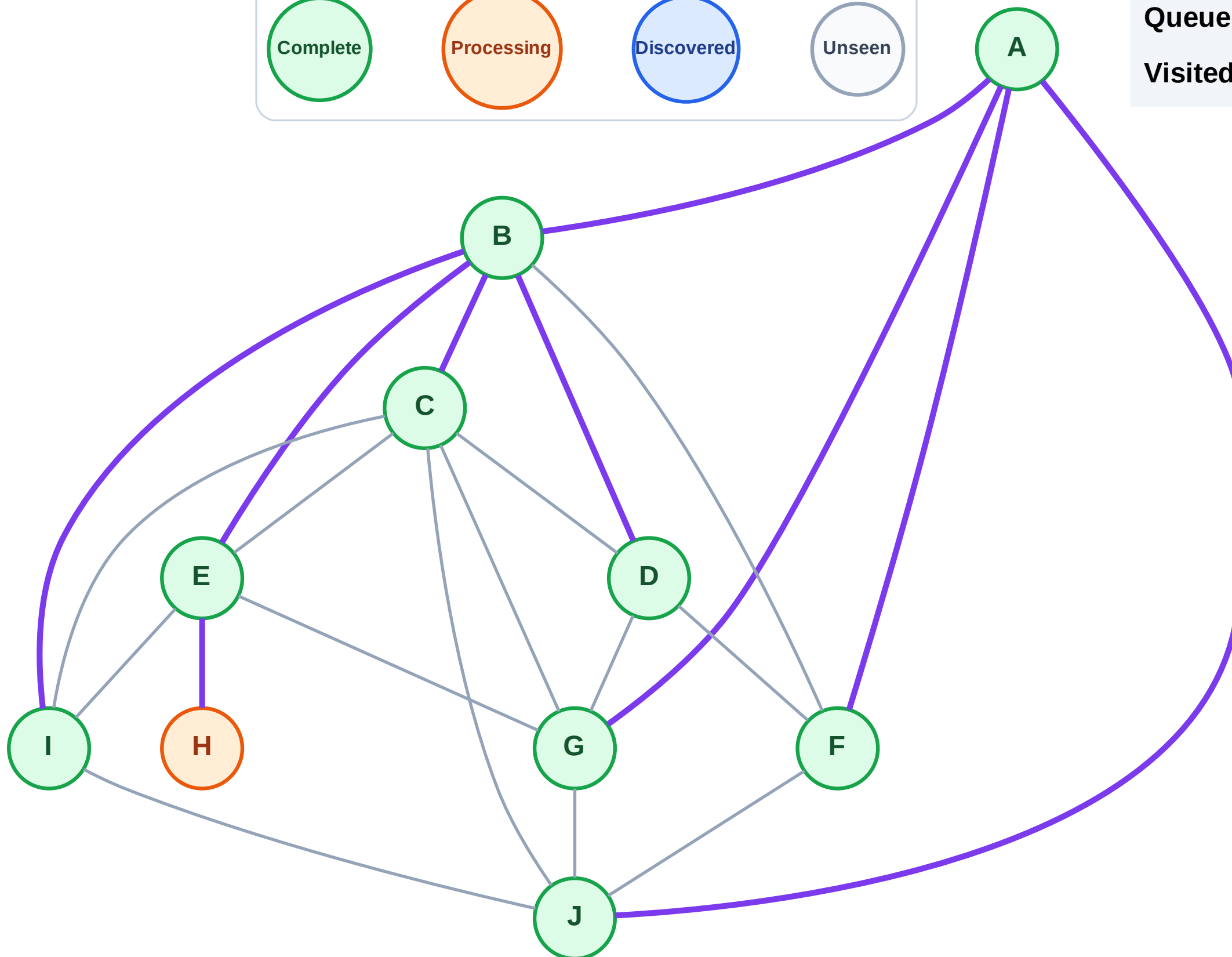
Discovered



Unseen

Queue: (empty)

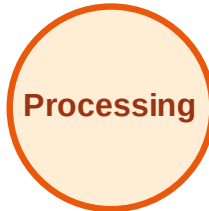
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

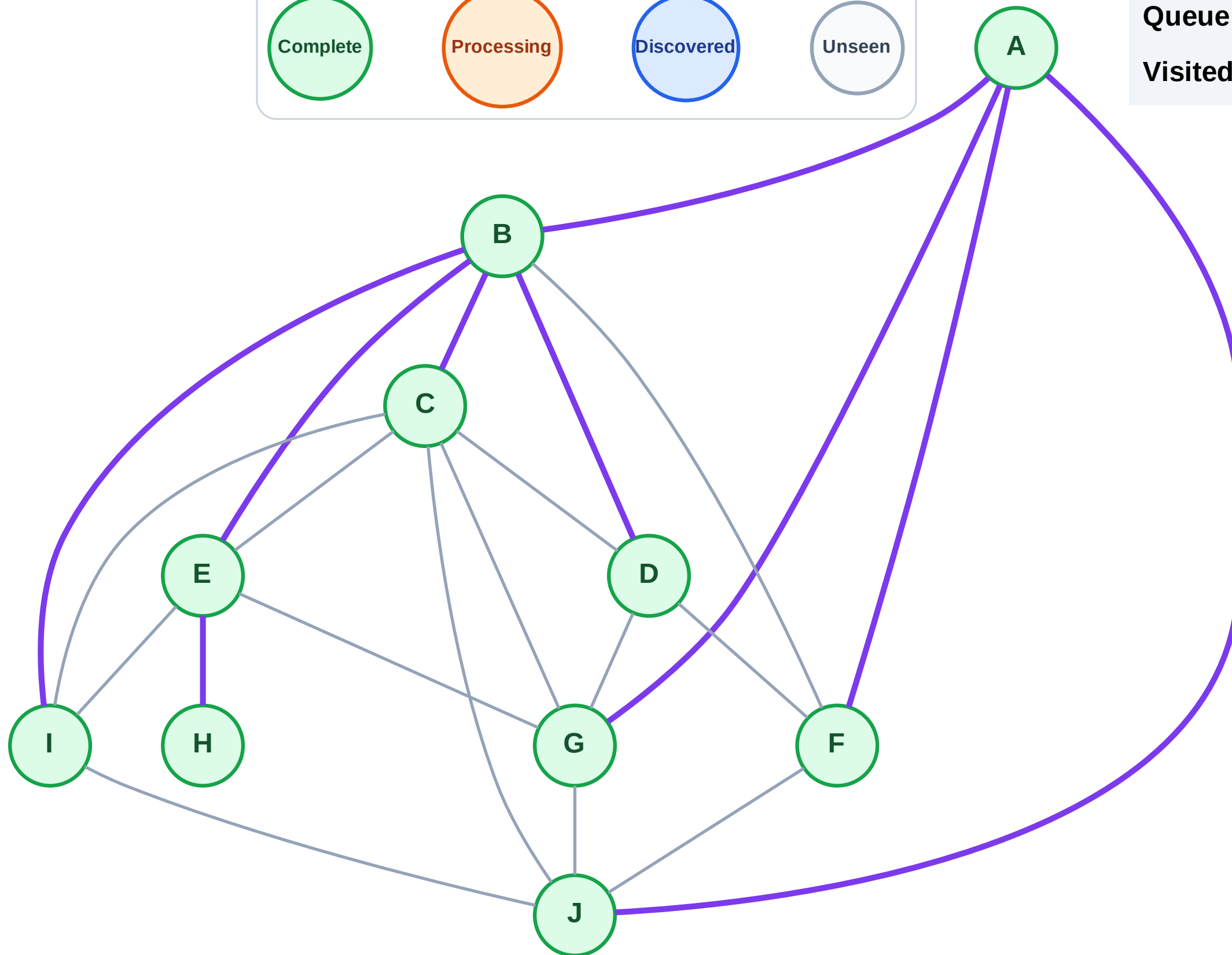
Step 21: No new neighbors from H

Legend



Queue: (empty)

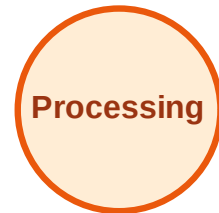
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

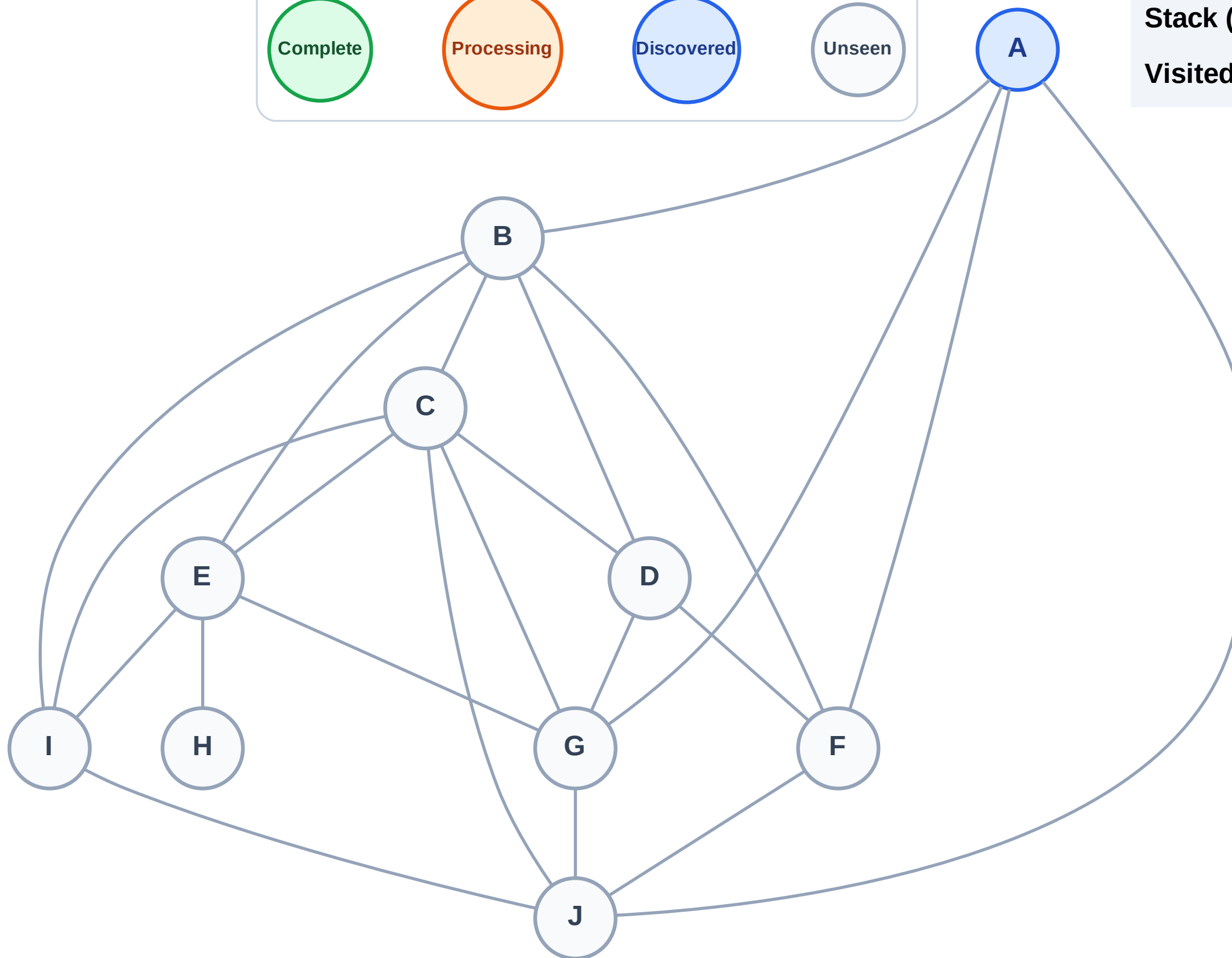
Step 1: Start at A

Legend



Stack (top at right): A

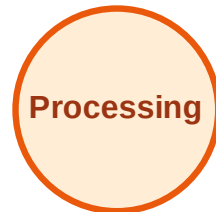
Visited: (none)



DFS Traversal

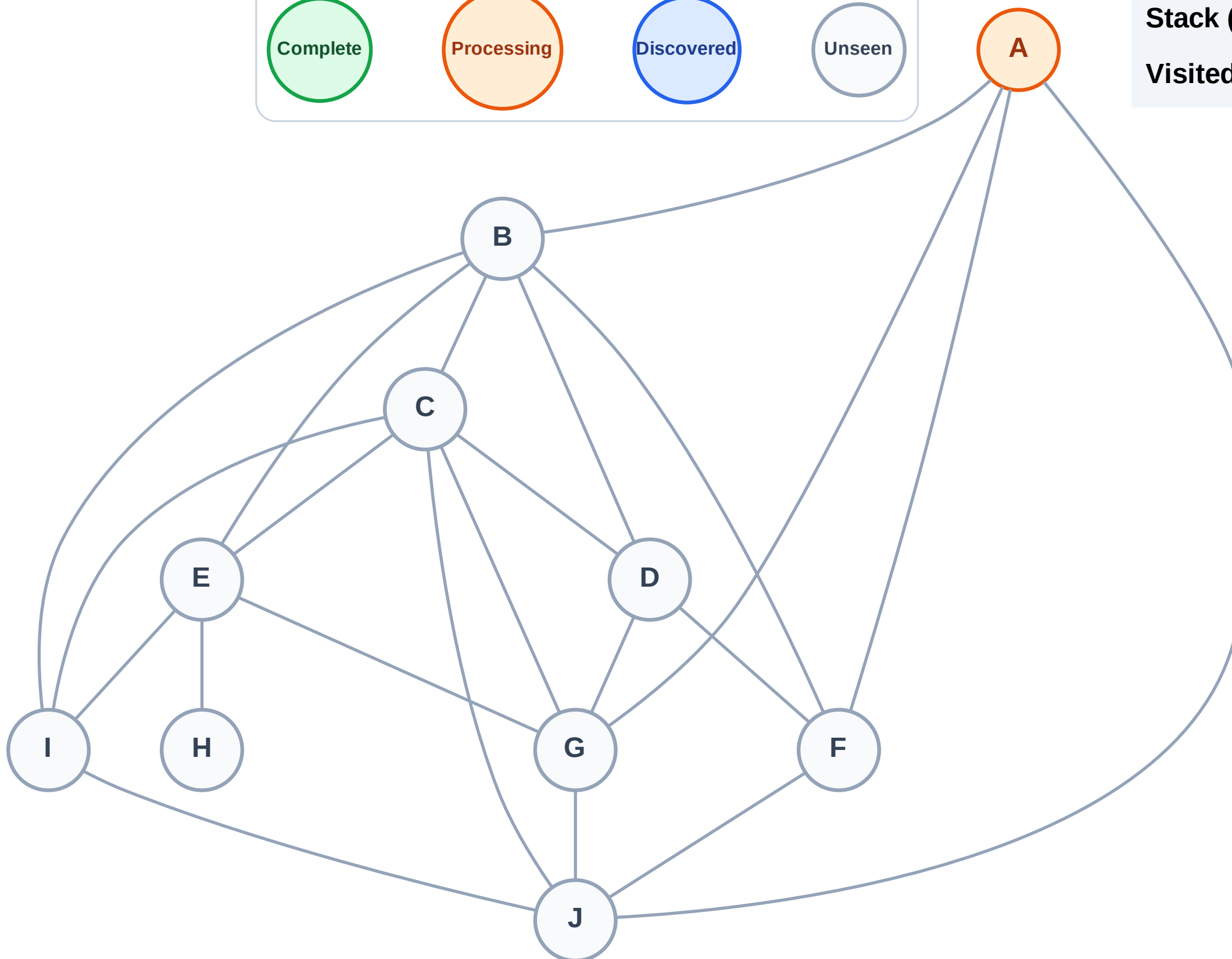
Step 2: Process node A

Legend



Stack (top at right): (empty)

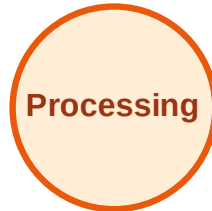
Visited: (none)



DFS Traversal

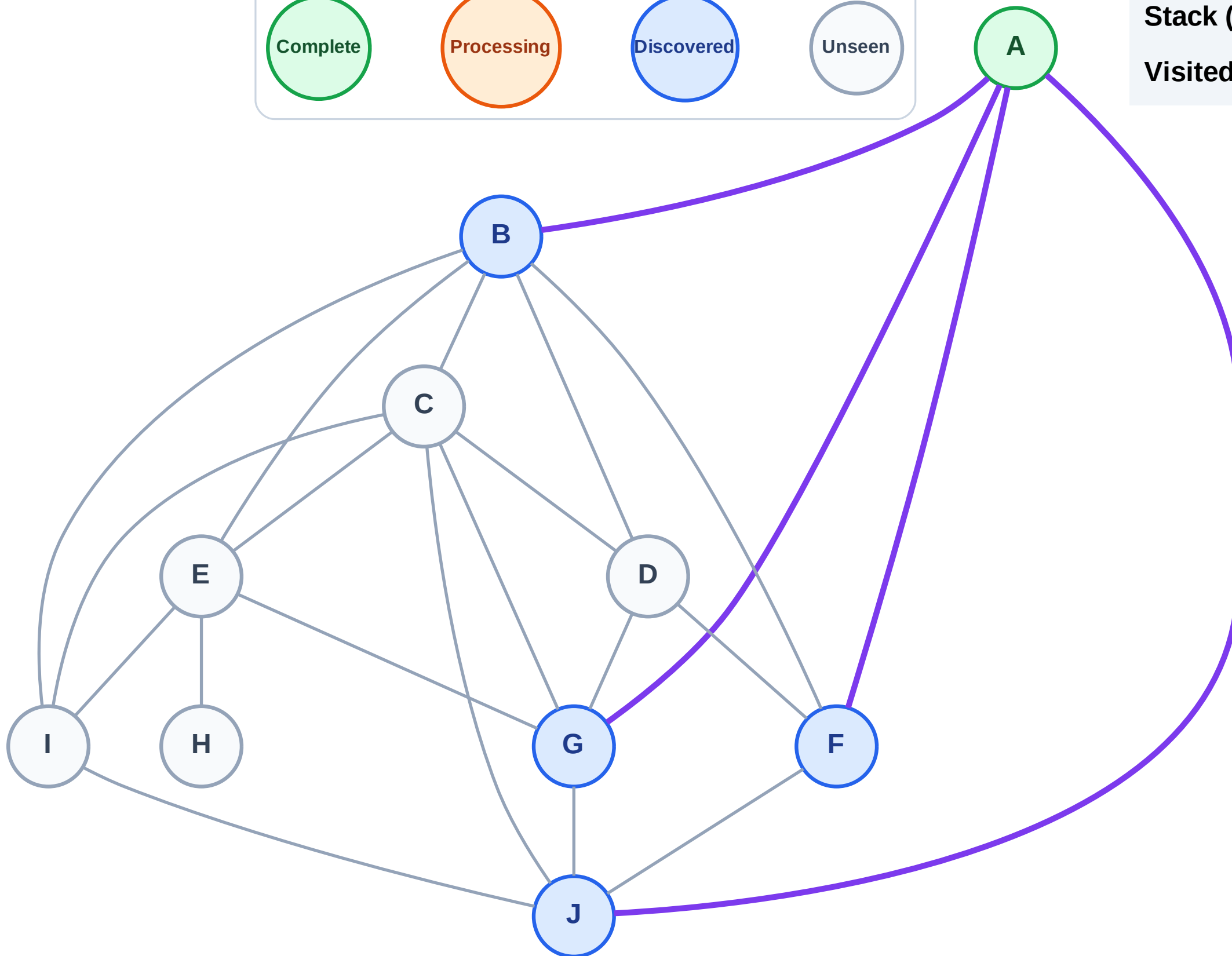
Step 3: Discover J, G, F, B from A

Legend



Stack (top at right): J | G | F | B

Visited: A



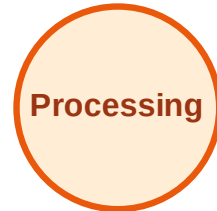
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



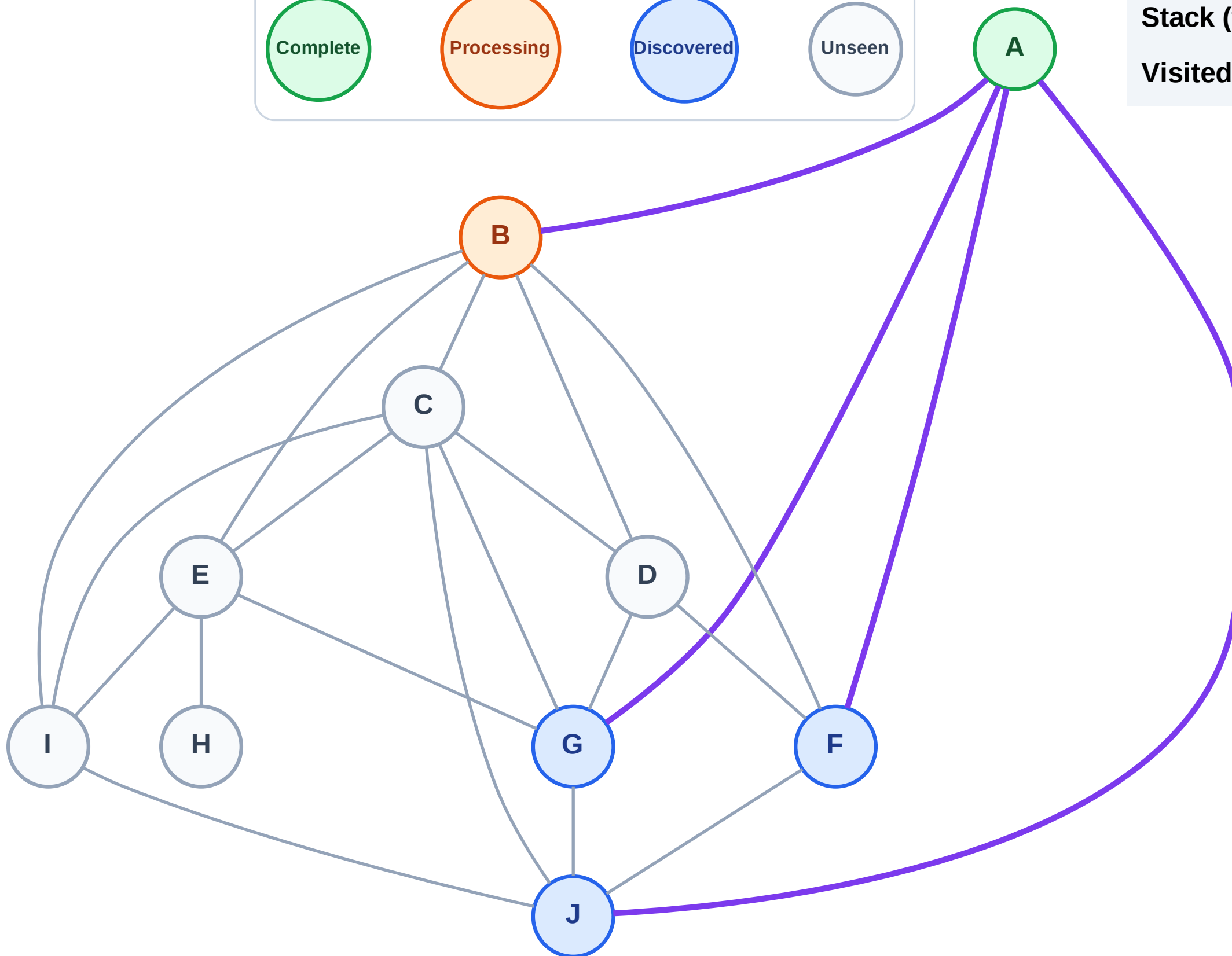
Discovered



Unseen

Stack (top at right): J | G | F

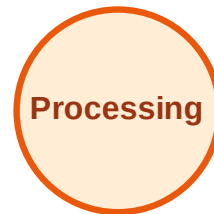
Visited: A



DFS Traversal

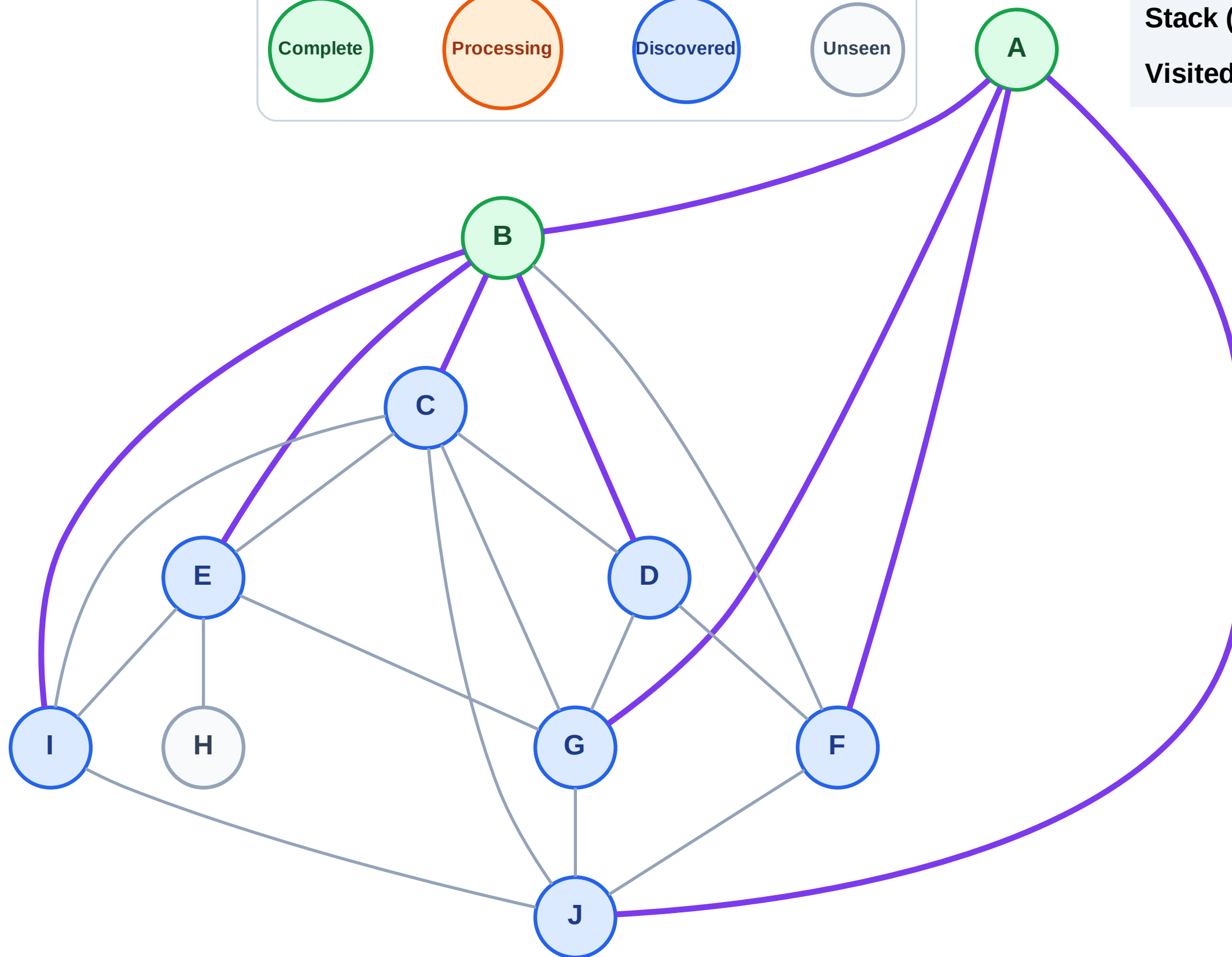
Step 5: Discover I, E, D, C from B

Legend



Stack (top at right): J | G | F | I | E | D | C

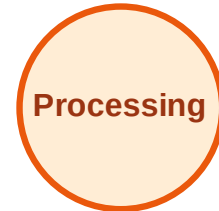
Visited: A, B



DFS Traversal

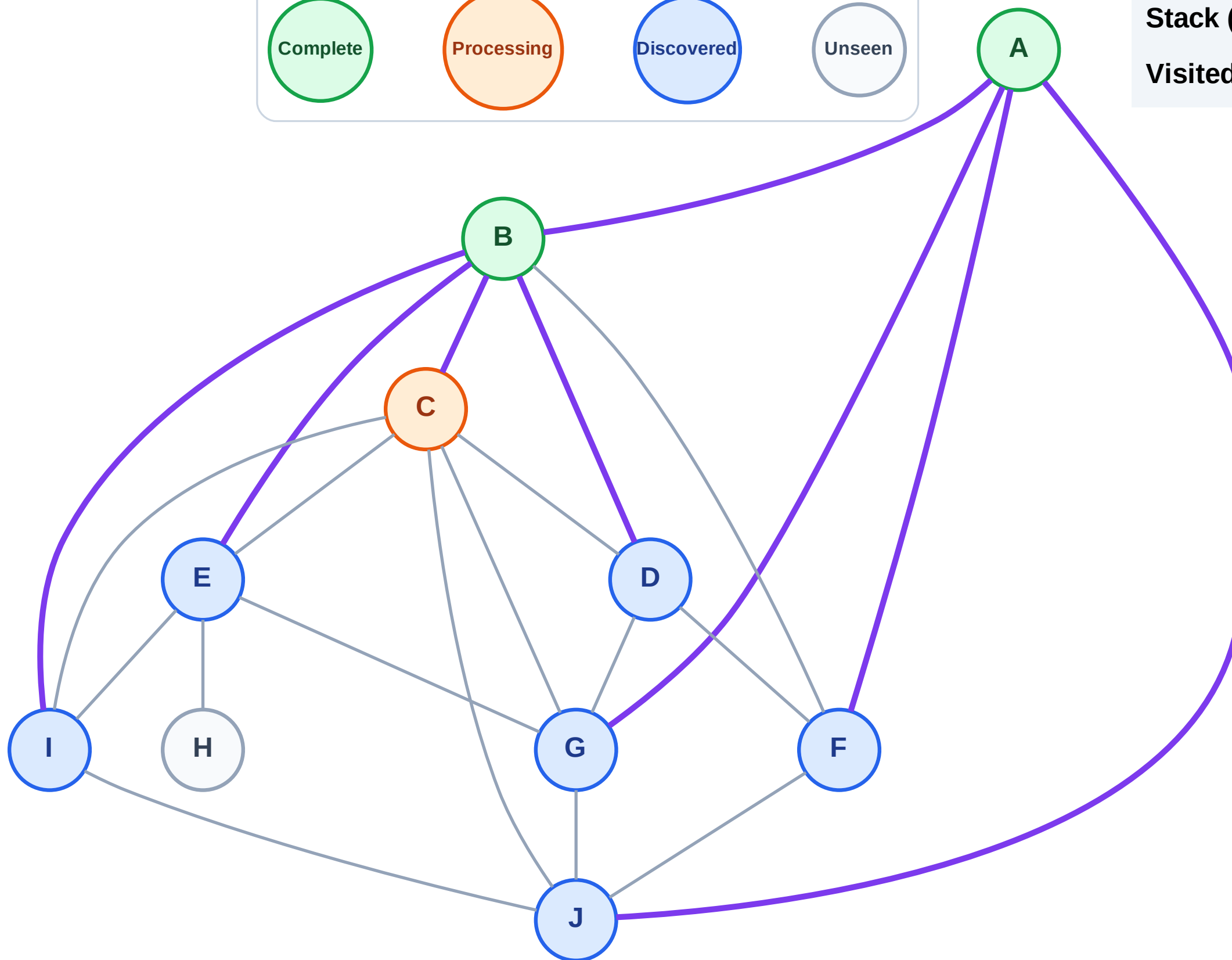
Step 6: Process node C

Legend



Stack (top at right): J | G | F | I | E | D

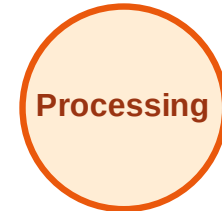
Visited: A, B



DFS Traversal

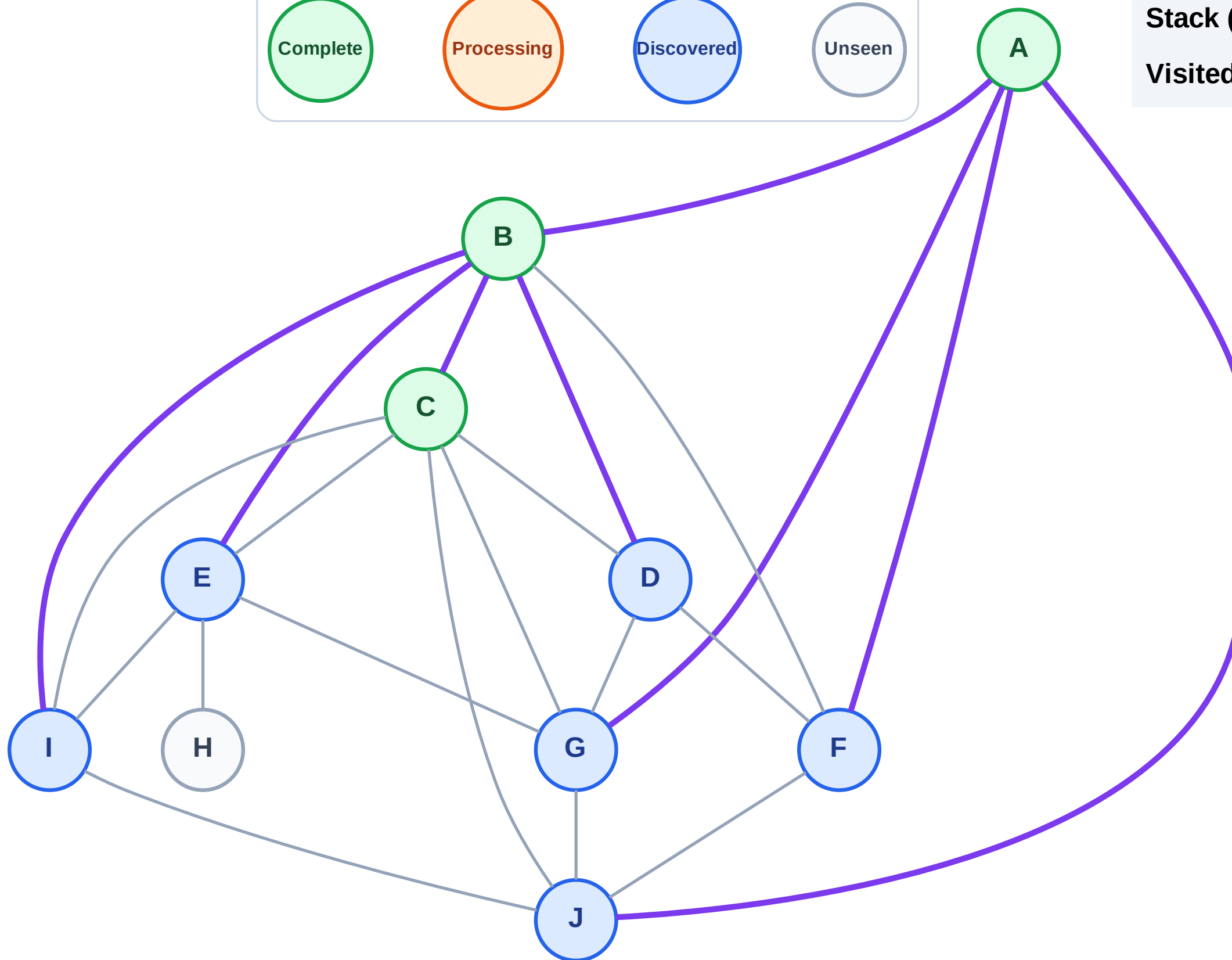
Step 7: No new neighbors from C

Legend



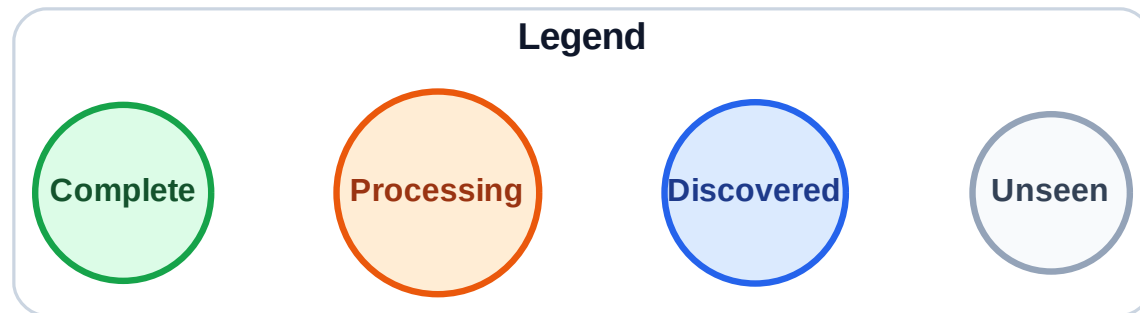
Stack (top at right): J | G | F | I | E | D

Visited: A, B, C



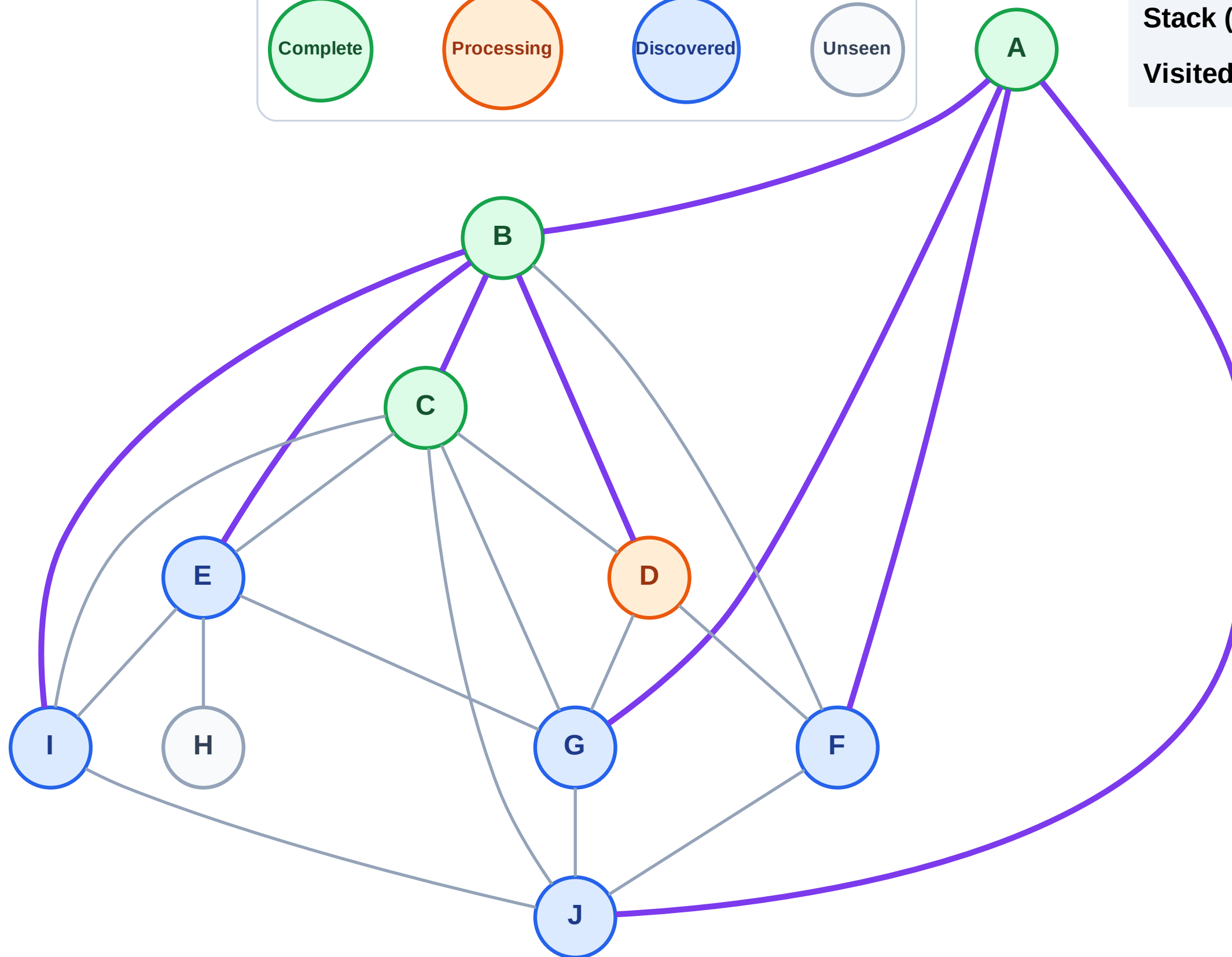
Step 8: Process node D

Step 8: Process node D



Stack (top at right): J | G | F | I | E

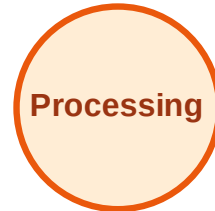
Visited: A, B, C



DFS Traversal

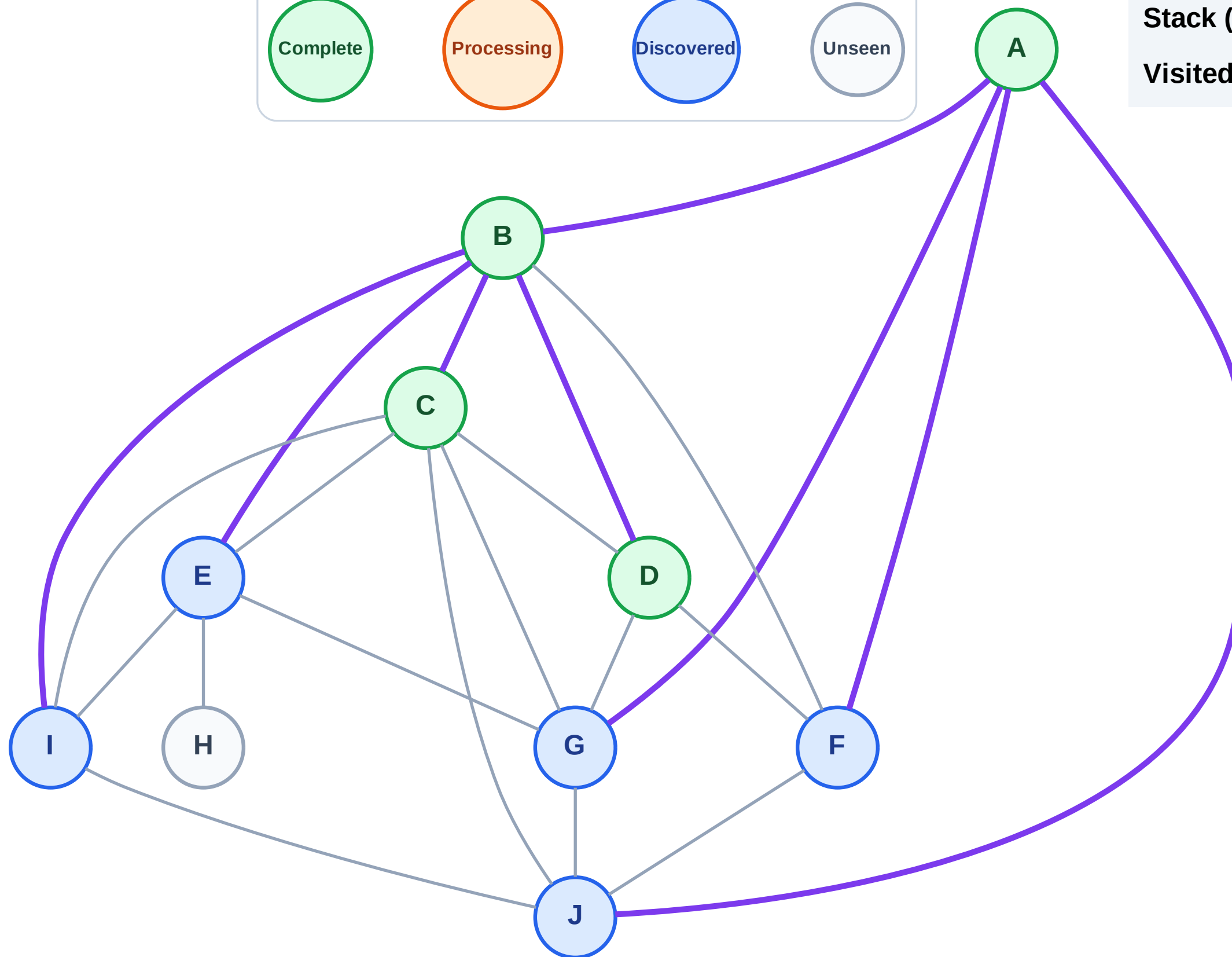
Step 9: No new neighbors from D

Legend



Stack (top at right): J | G | F | I | E

Visited: A, B, C, D



DFS Traversal

Step 10: Process node E

Legend

Complete

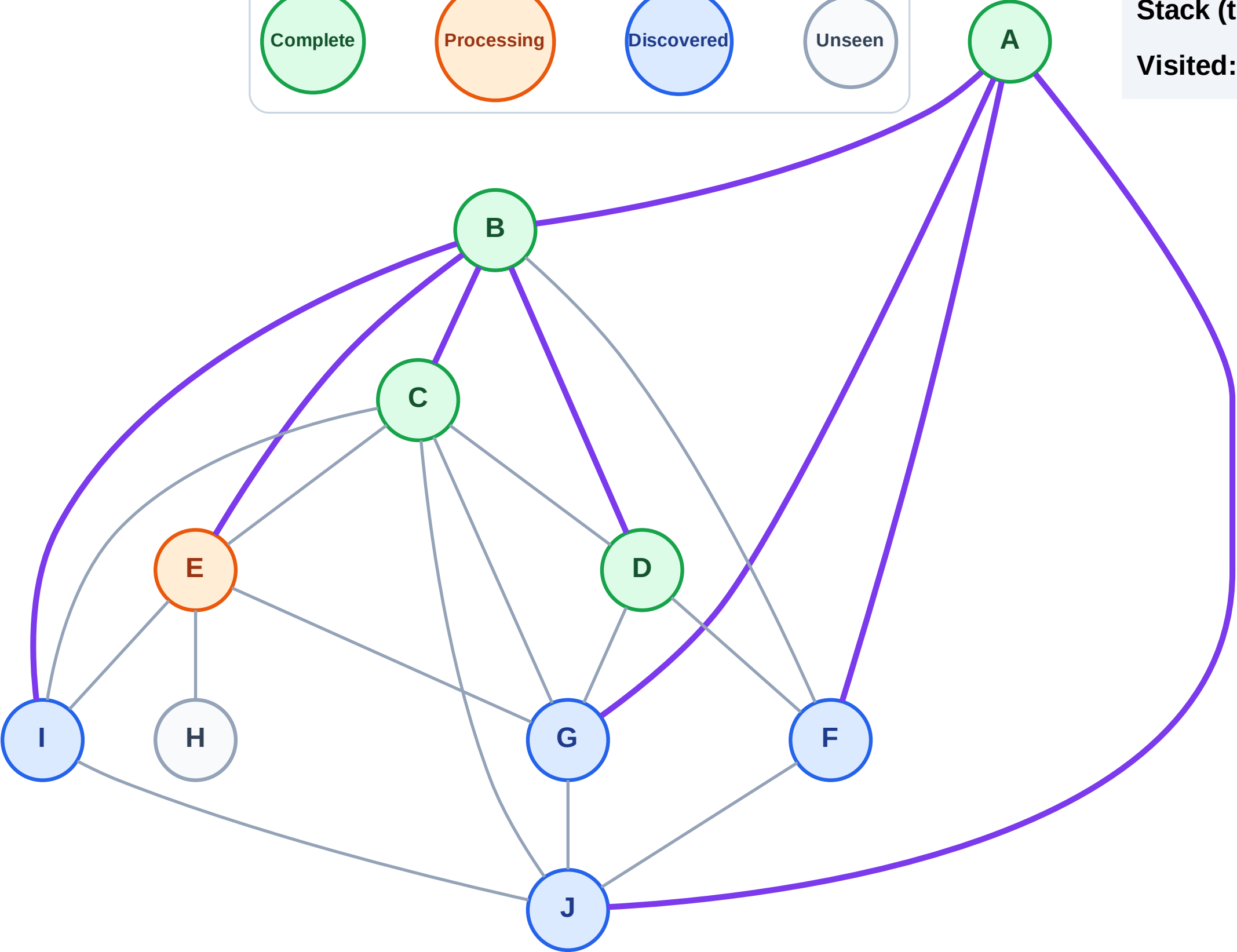
Processing

Discovered

Unseen

Stack (top at right): J | G | F | I

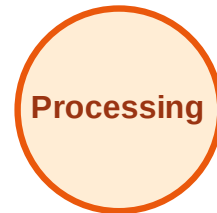
Visited: A, B, C, D



DFS Traversal

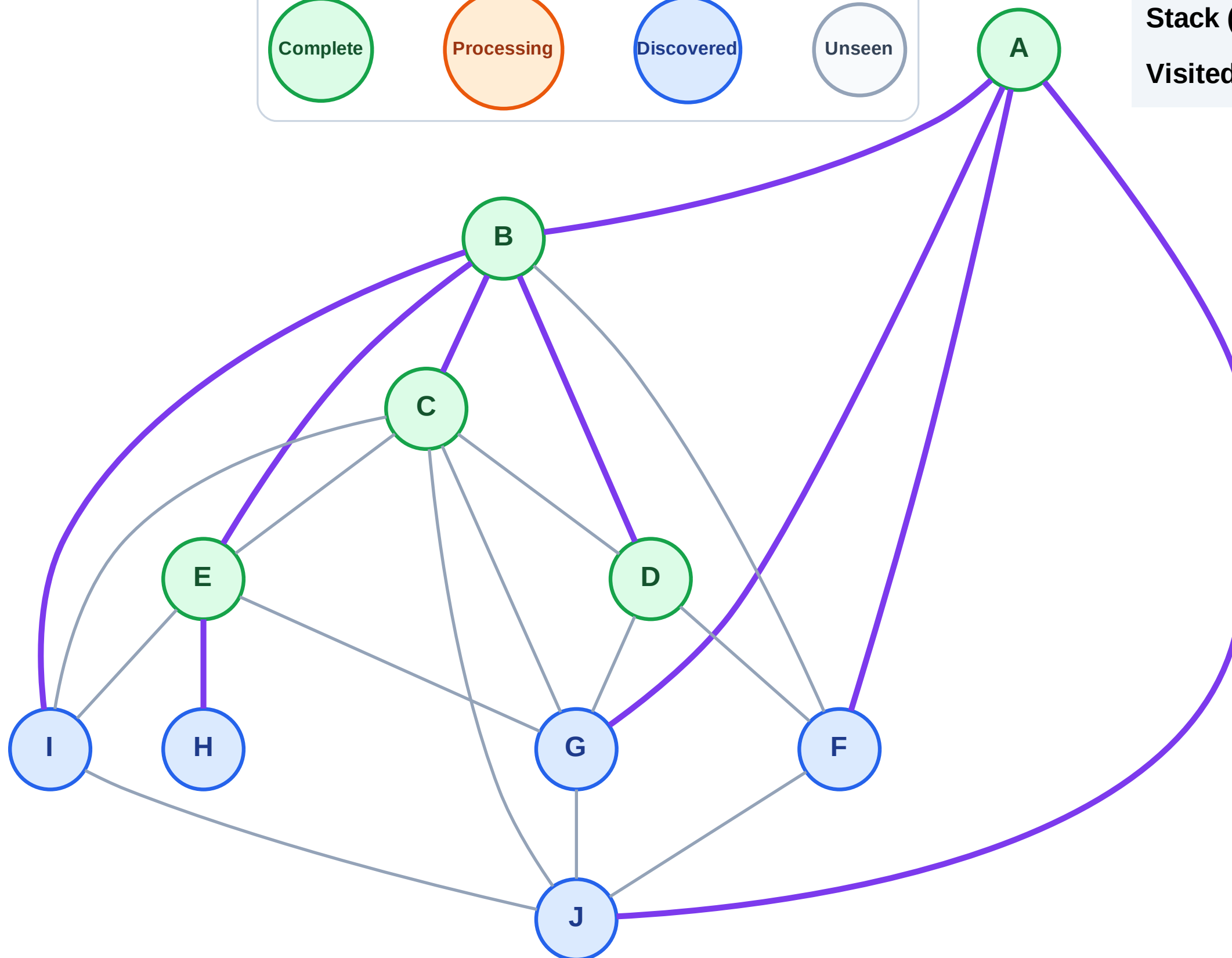
Step 11: Discover H from E

Legend



Stack (top at right): J | G | F | I | H

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node H

Legend

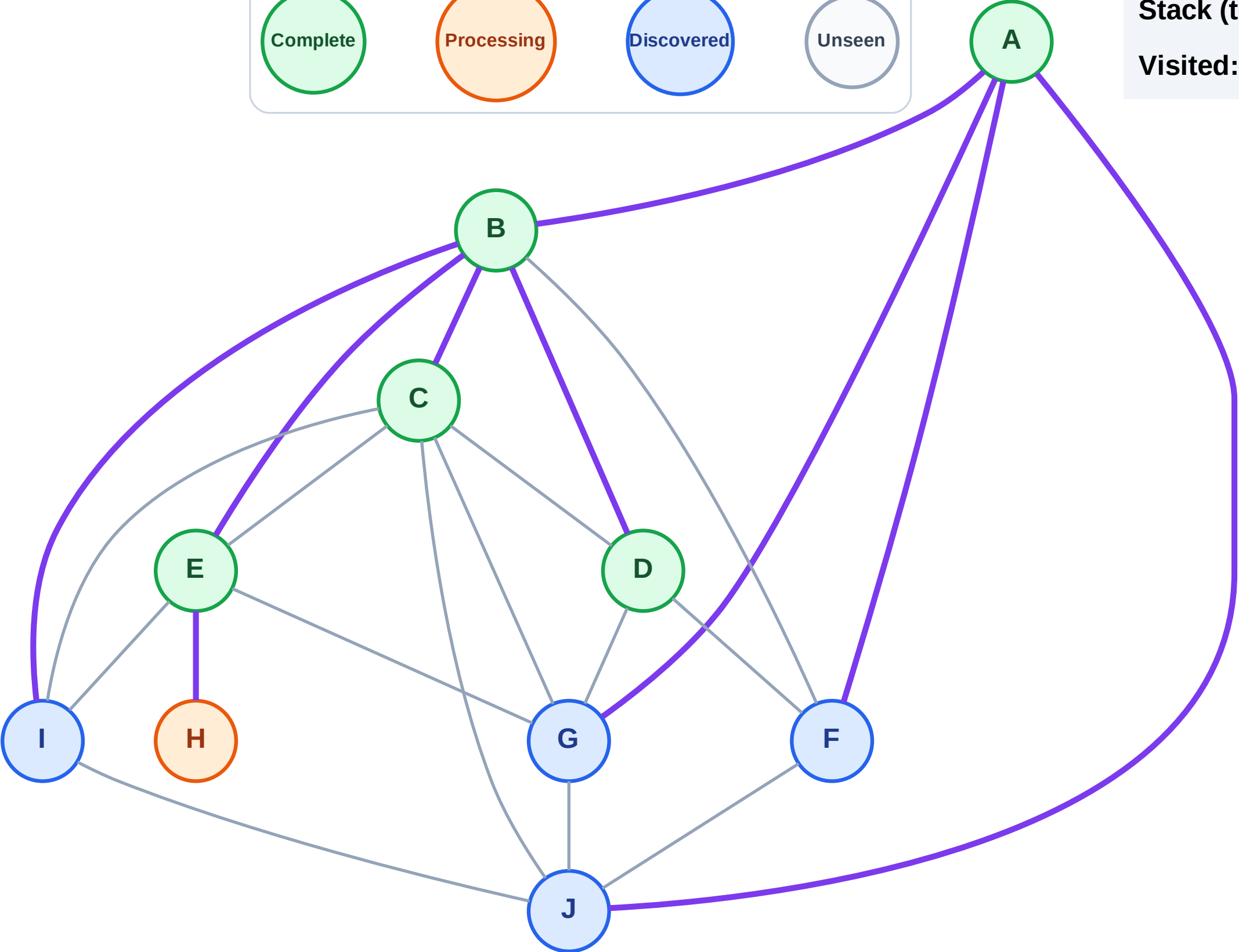
Complete

Processing

Discovered

Unseen

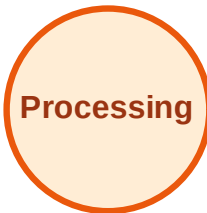
Stack (top at right): J | G | F | I
Visited: A, B, C, D, E



DFS Traversal

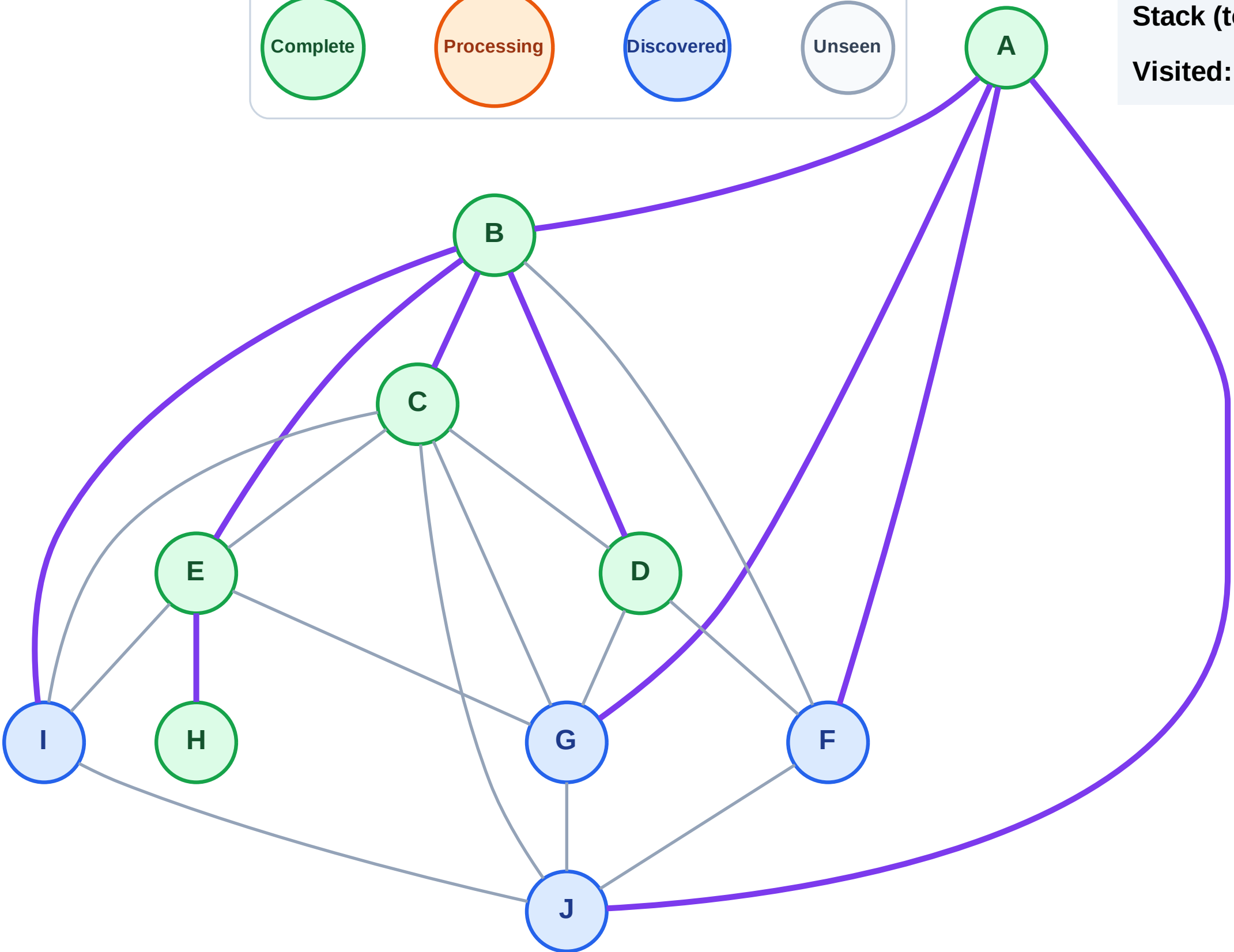
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F | I

Visited: A, B, C, D, E, H



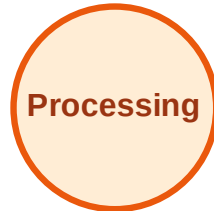
DFS Traversal

Step 14: Process node I

Legend



Complete



Processing



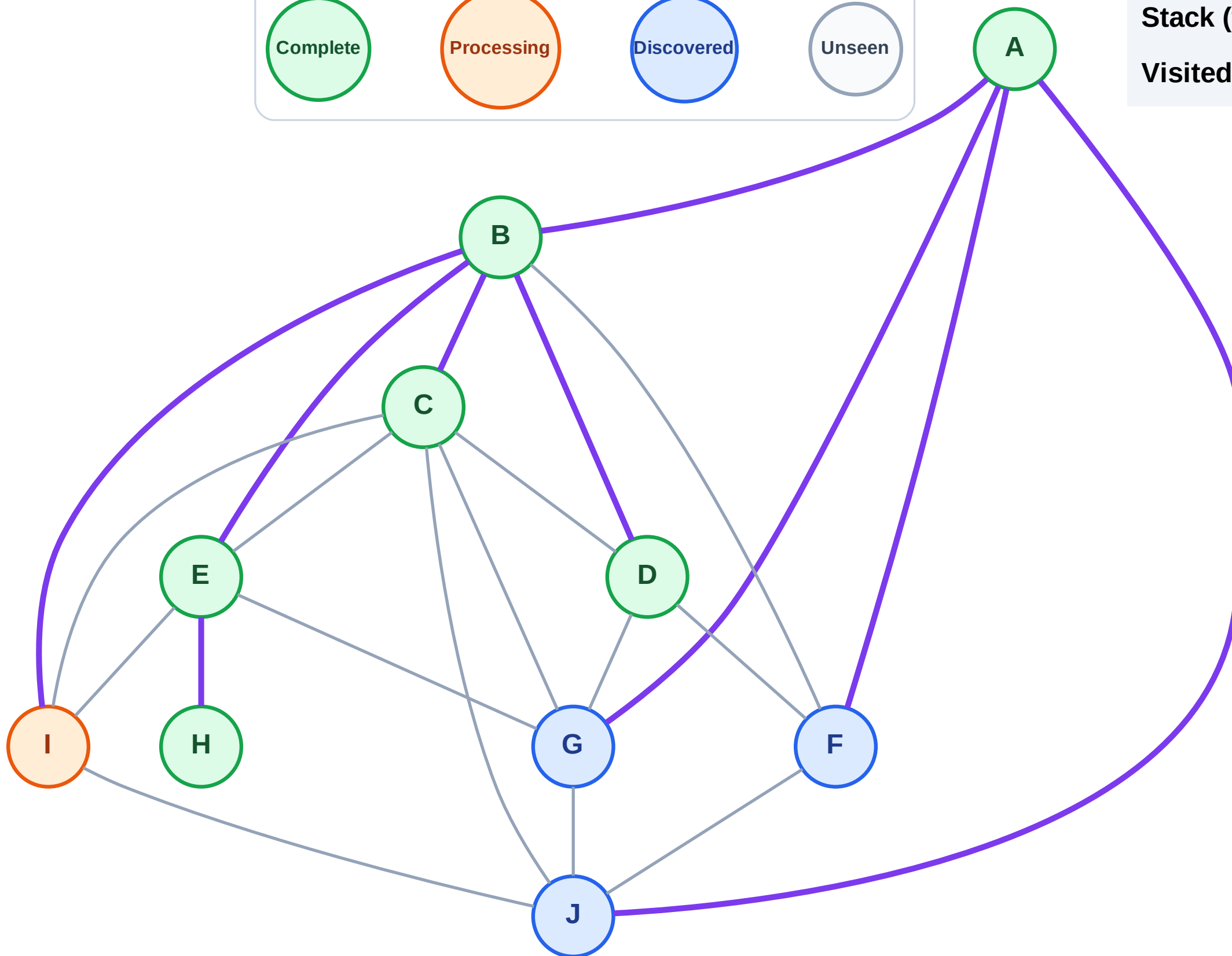
Discovered



Unseen

Stack (top at right): J | G | F

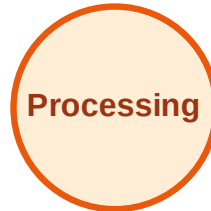
Visited: A, B, C, D, E, H



DFS Traversal

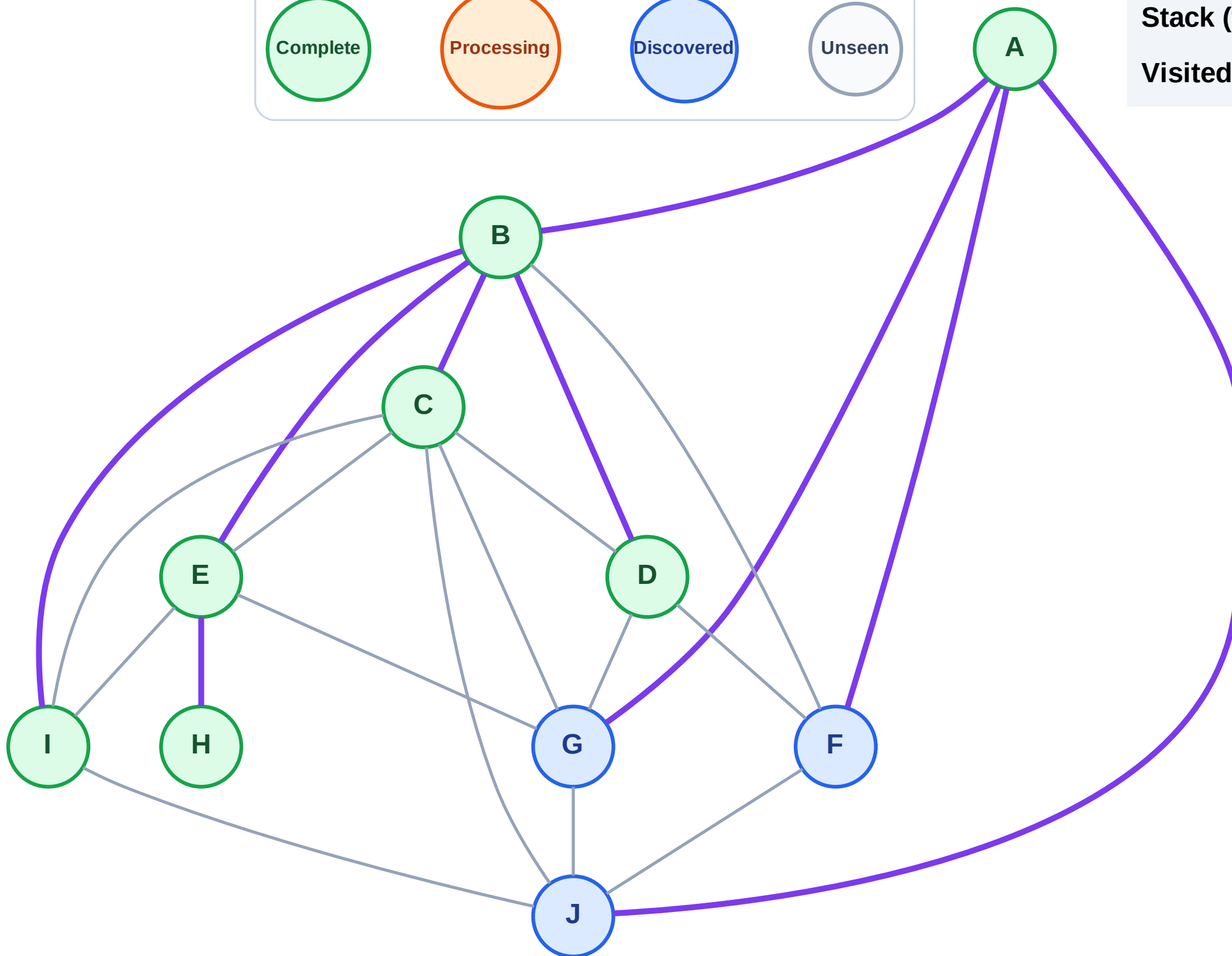
Step 15: No new neighbors from I

Legend



Stack (top at right): J | G | F

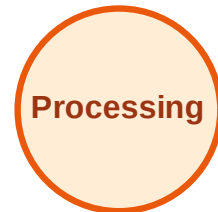
Visited: A, B, C, D, E, H, I



DFS Traversal

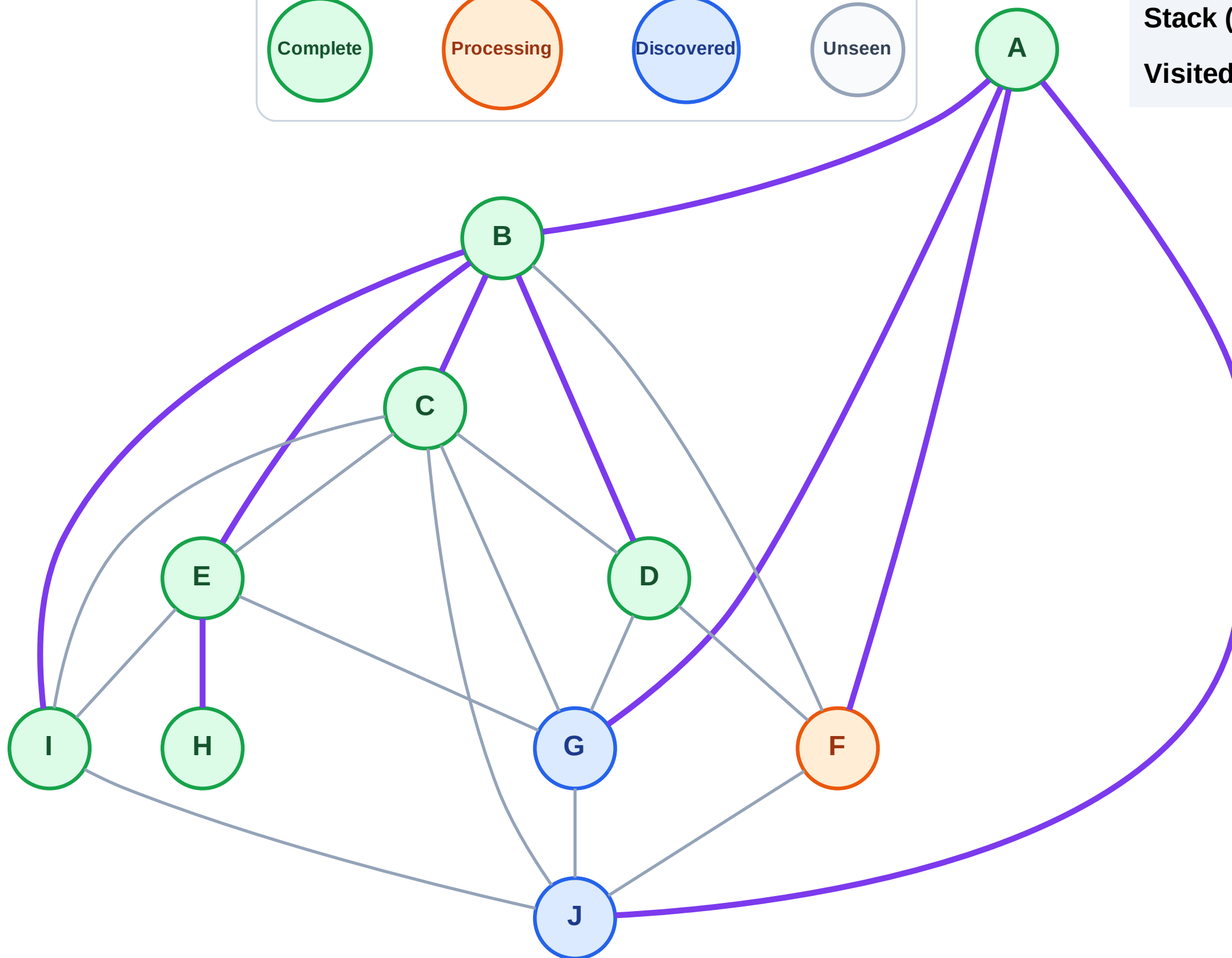
Step 16: Process node F

Legend



Stack (top at right): J | G

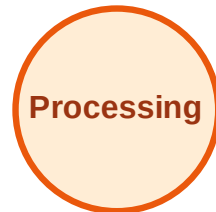
Visited: A, B, C, D, E, H, I



DFS Traversal

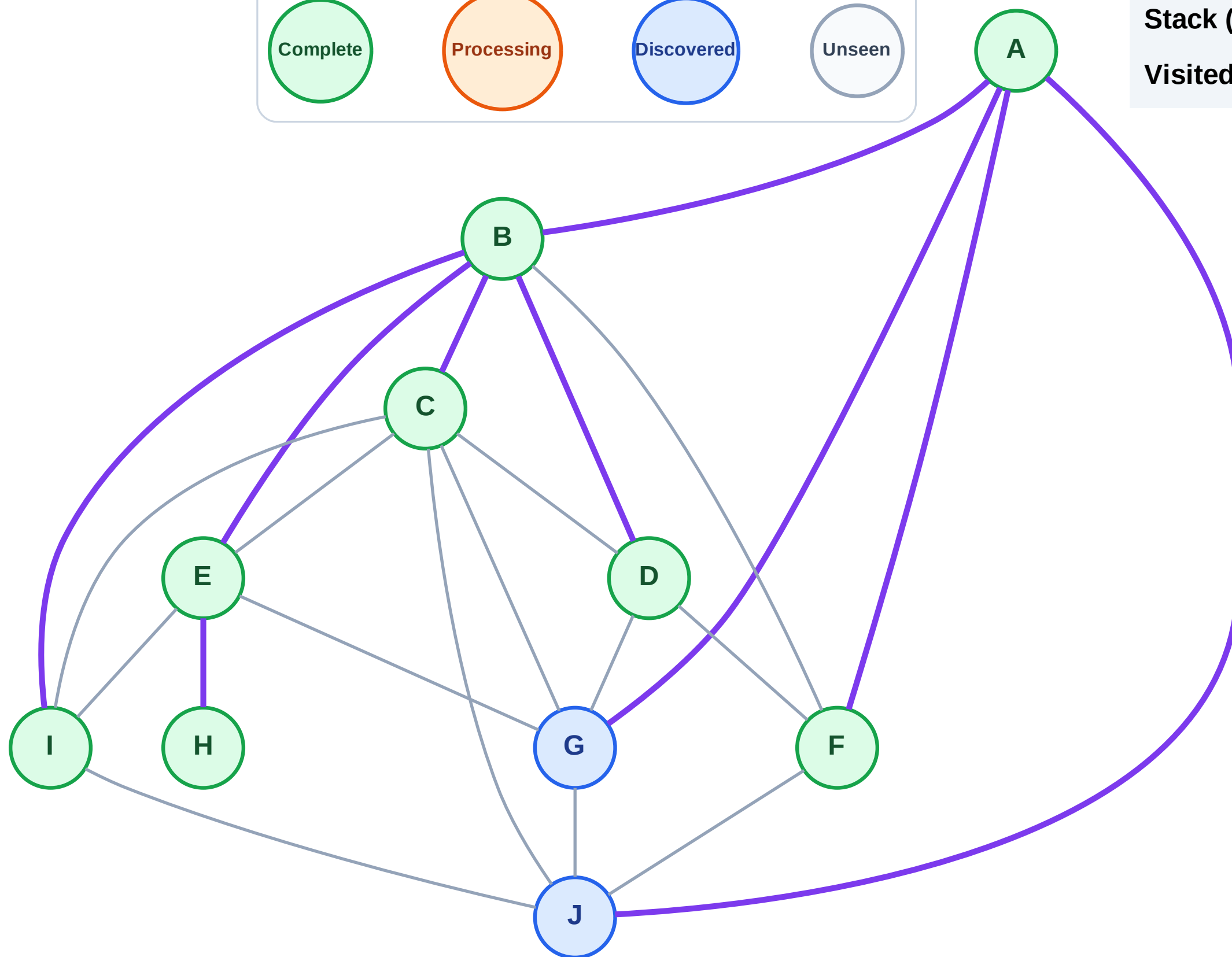
Step 17: No new neighbors from F

Legend



Stack (top at right): J | G

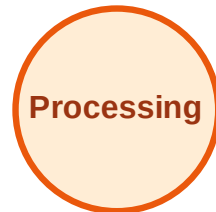
Visited: A, B, C, D, E, F, H, I



DFS Traversal

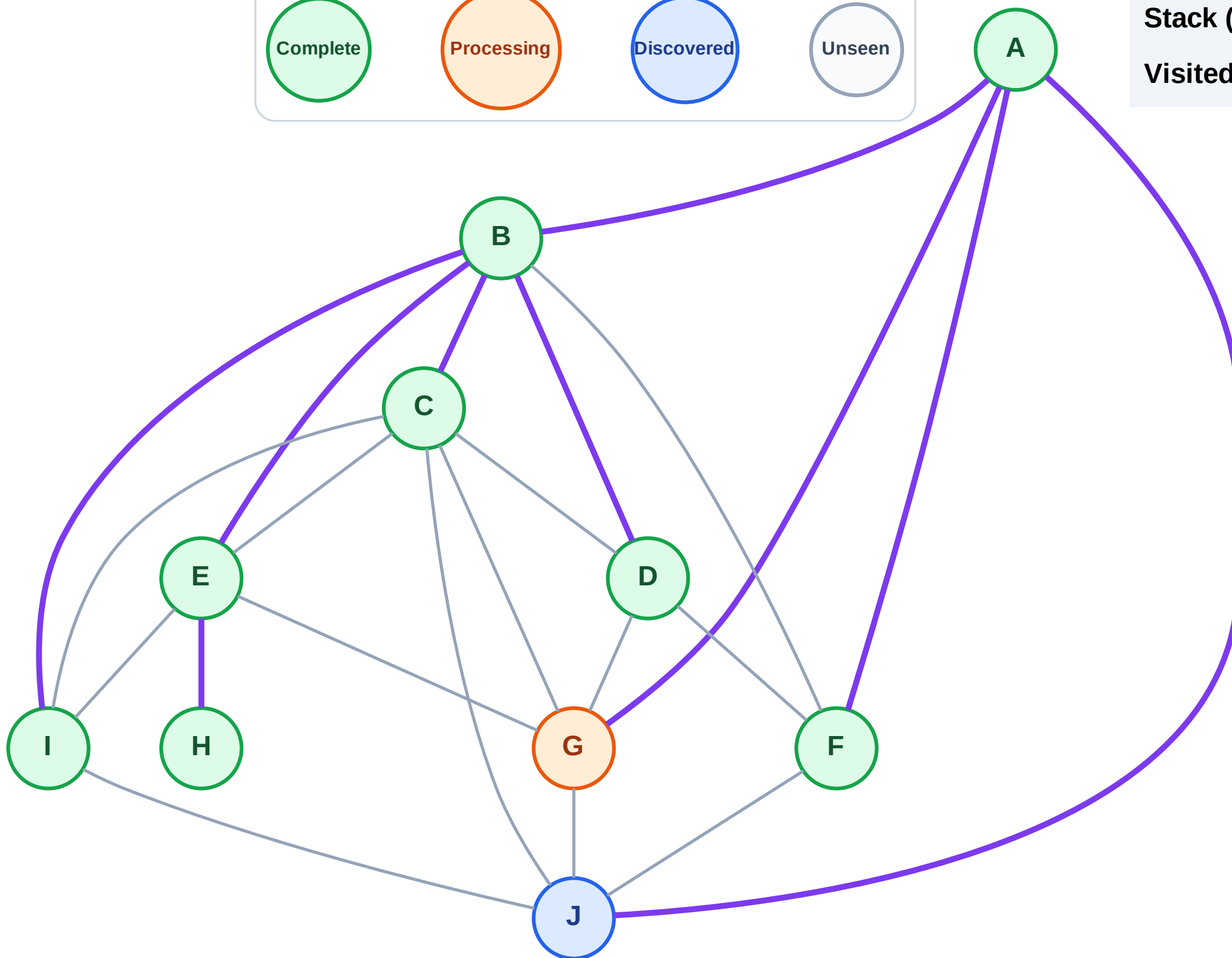
Step 18: Process node G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, H, I



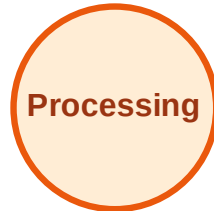
DFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



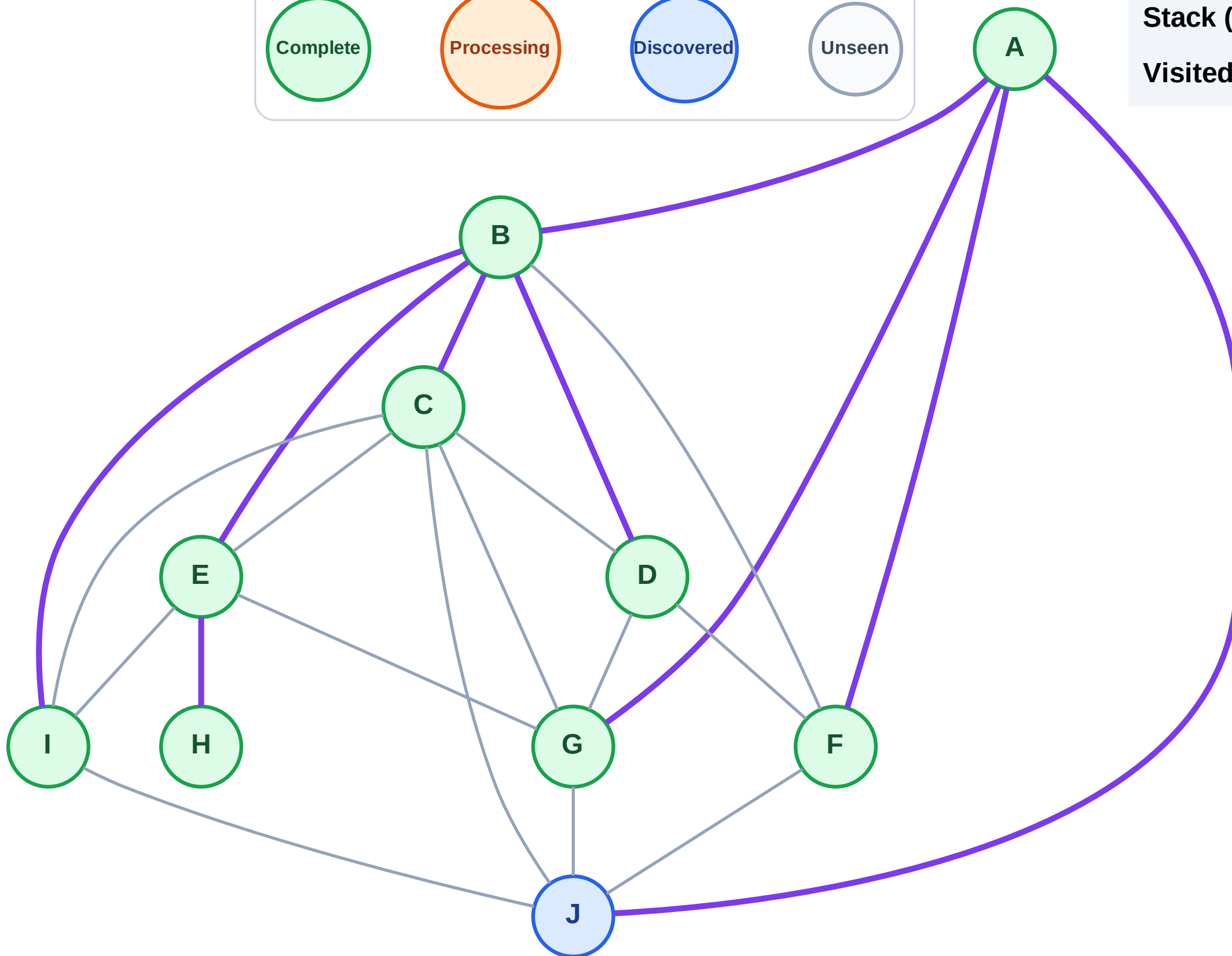
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



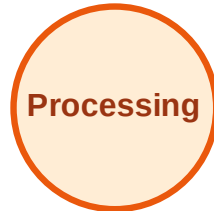
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



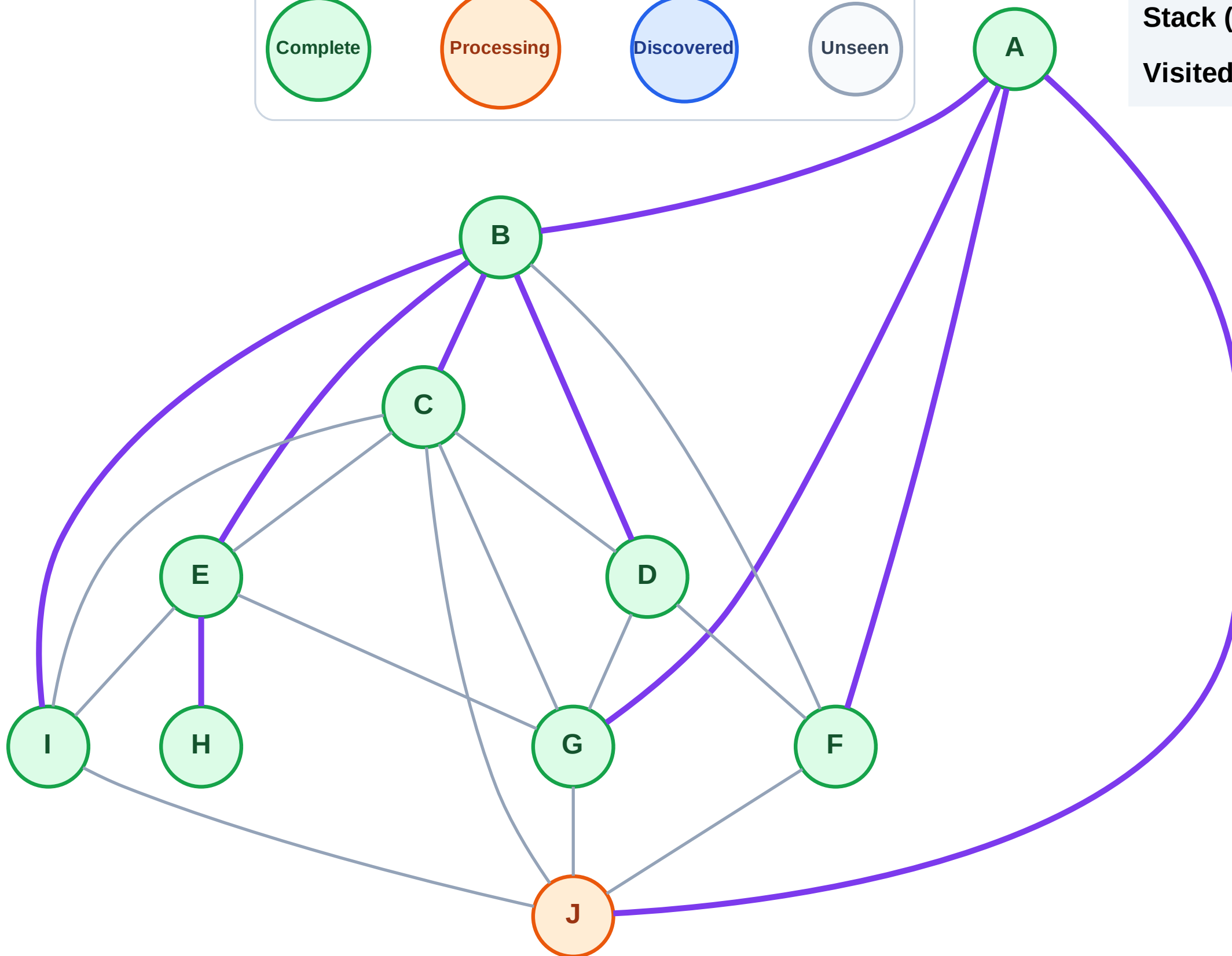
Discovered



Unseen

Stack (top at right): (empty)

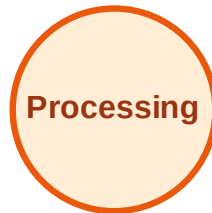
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

